Kubernetes Practice

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Korea Polytechnics

2026

KUBERNETES 개요

DEFINITION

컨테이너화된 애플리케이션의 배포·확장·운영을 자동화하는 오픈소스 컨테이너 오케스트레이션 플랫폼

NECESSITY

컨테이너 수 급증에 따른 배포 관리, 장애 복구, 자동 확장 및 네트워크 운영 자동화 필요성 대두

CORE FUNCTIONS

- 자동 배포 & Self-Healing
- Scaling & Load Balancing
- Service Discovery
- Rolling Update

CLUSTER STRUCTURE

Control Plane

API Server, Scheduler, etcd

Worker Node

Pod (Container) 실행 환경

HIERARCHY

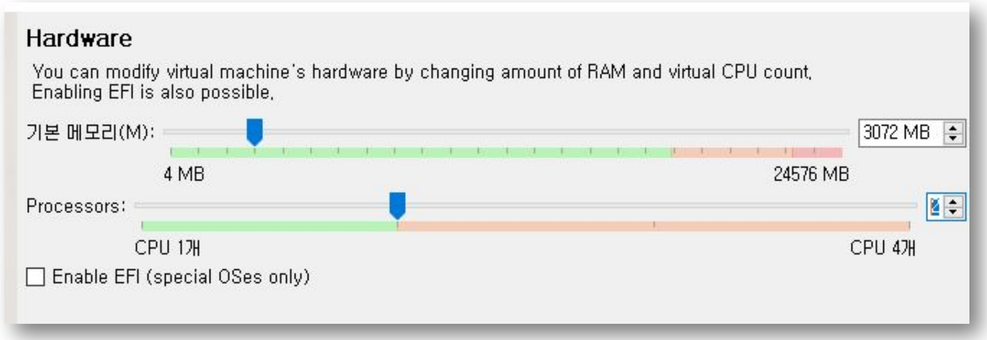
Cluster › Node › Pod

CORE SUMMARY

여러 서버에 분산된 컨테이너를 하나의 클러스터로 묶어 자동으로 배포·확장·복구·관리하는 핵심 플랫폼입니다.

Requirements

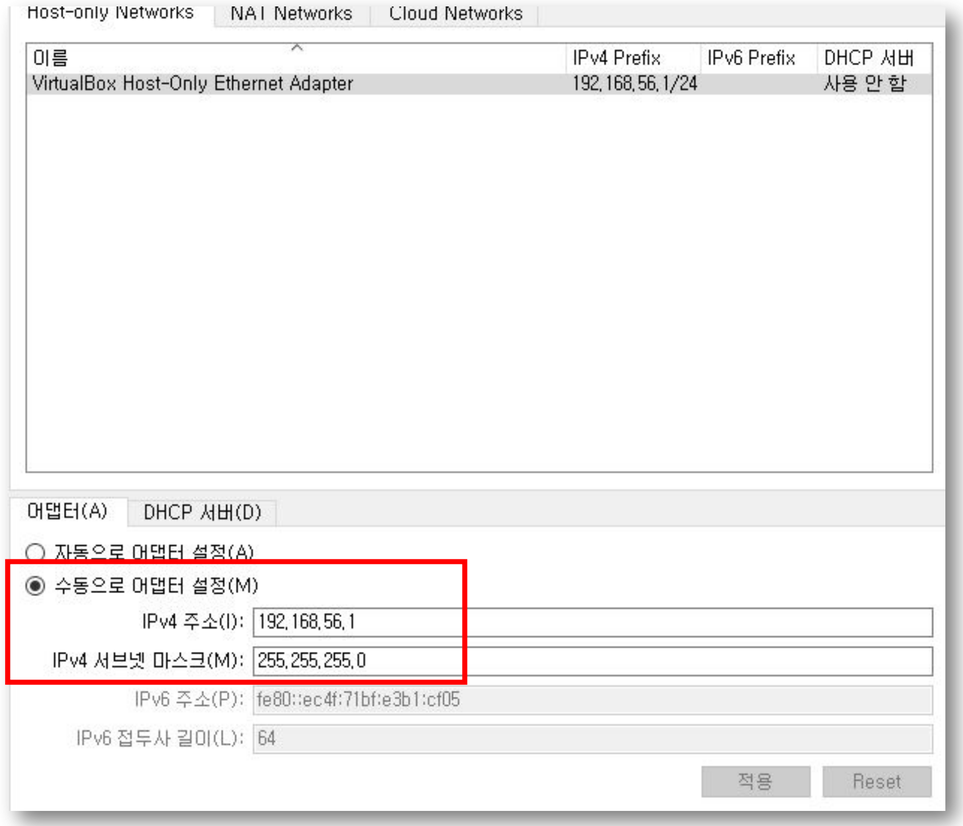
- We’ll make 2 VMs, one is master node and the other is worker node. Each node need to be set as below
- CPU : 2 core (minimum)
- RAM : 3GB (minimum)
- Storage: 30GB(minimum)
- OS : Ubuntu 24.04 (preferred)

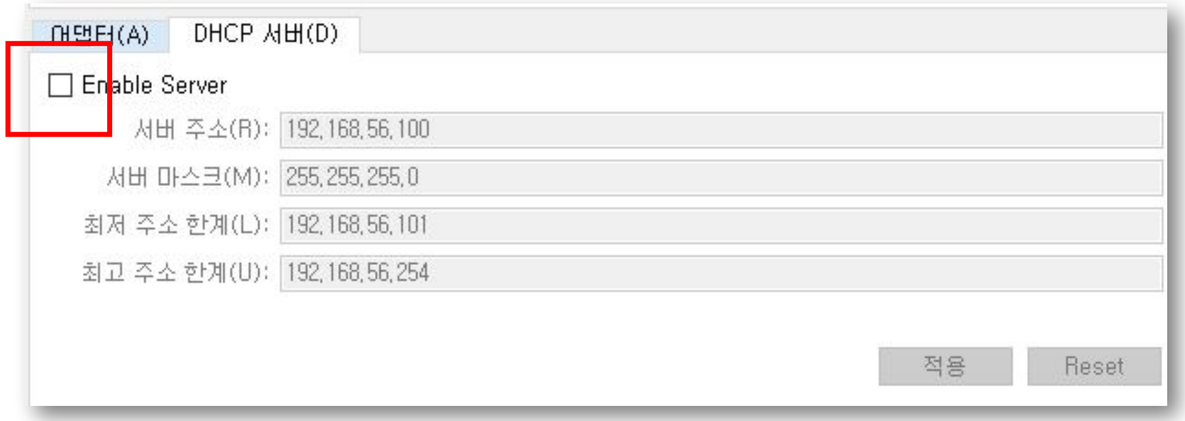


Network setup

In virtual box

- 1. Set a host network manager (menu -> file -> host network manager…)

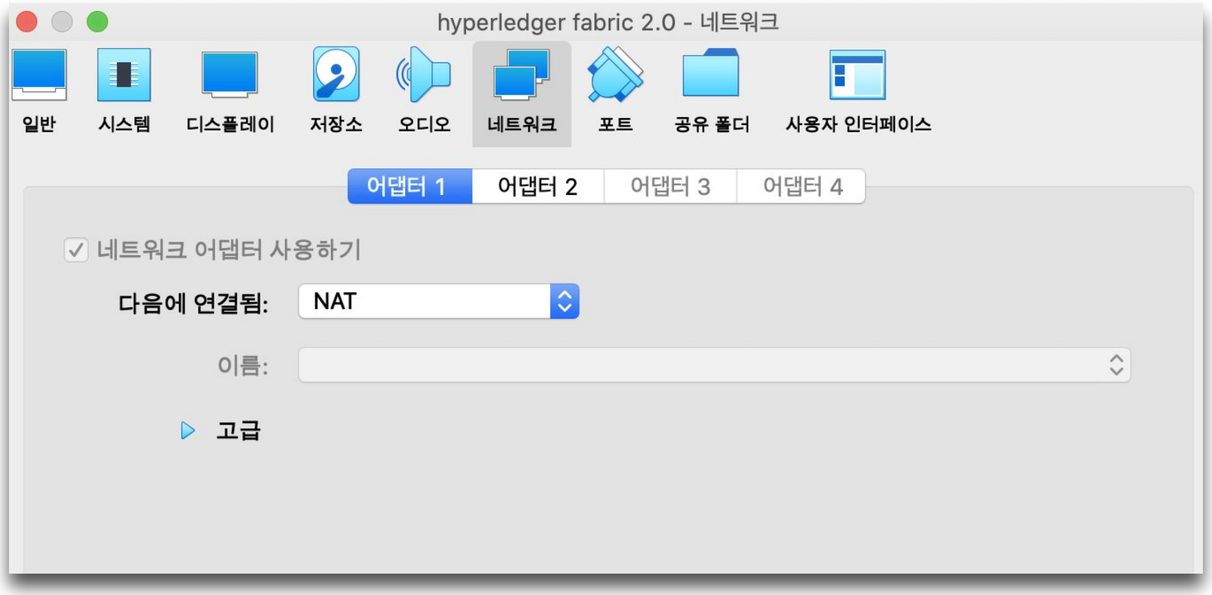


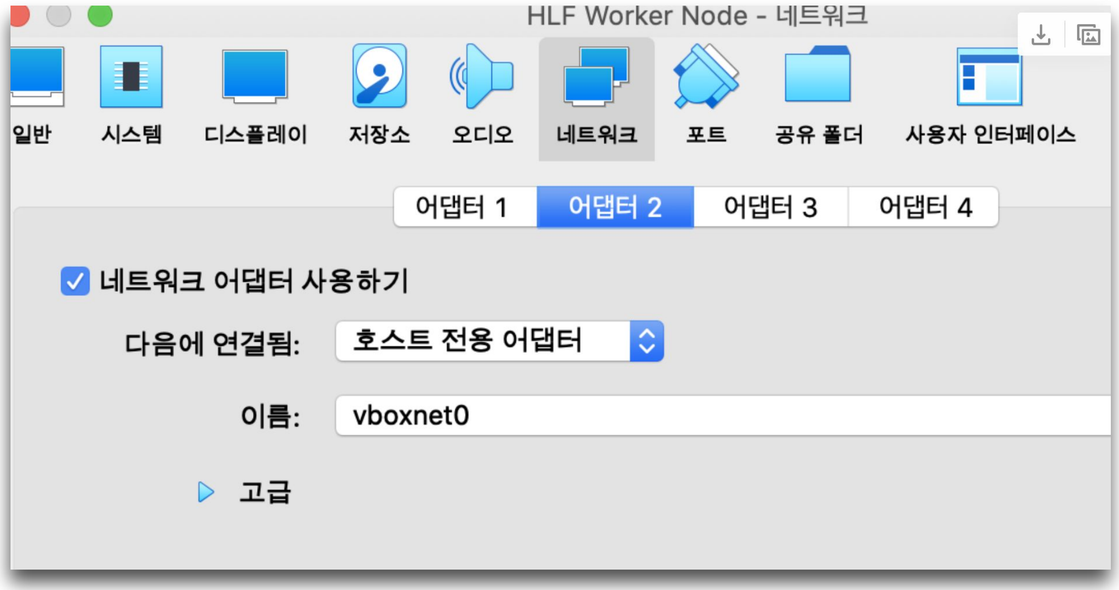


Network setup

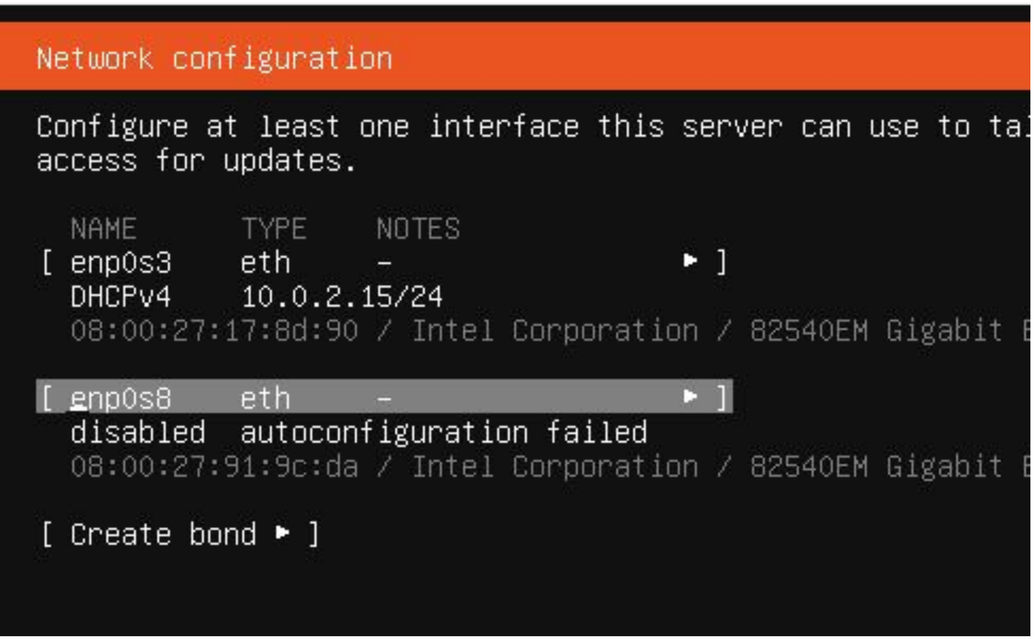
In virtual box

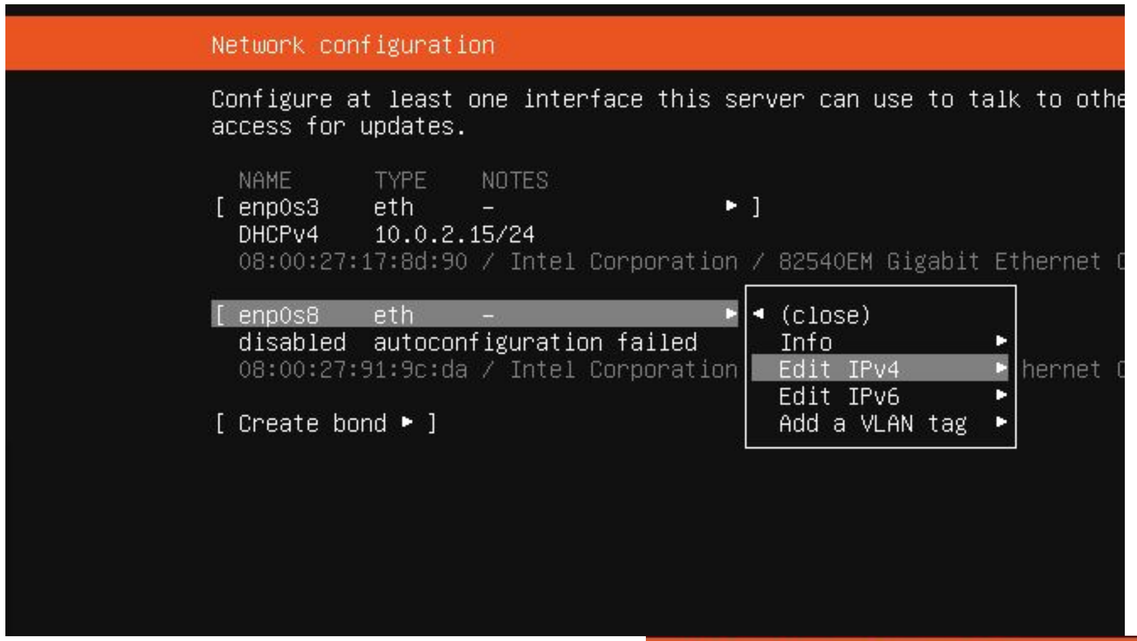
- Adaptor 1 : NAT
- Adaptor 2: Host-Only Network

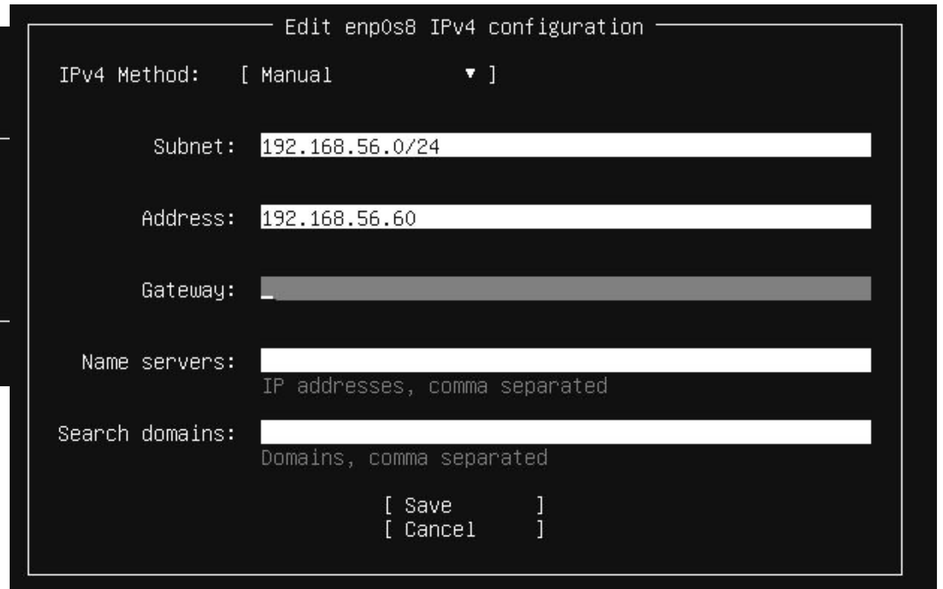


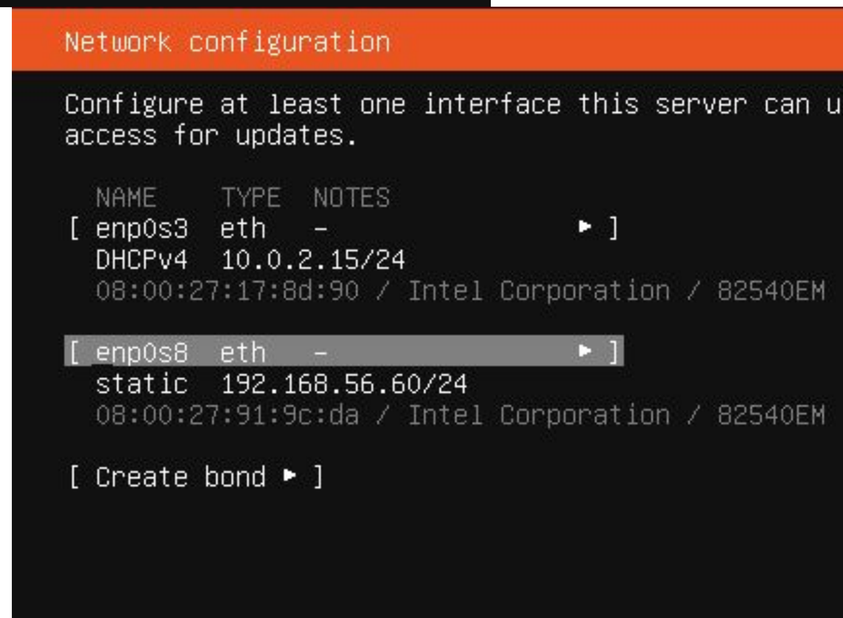


setup network info while installing









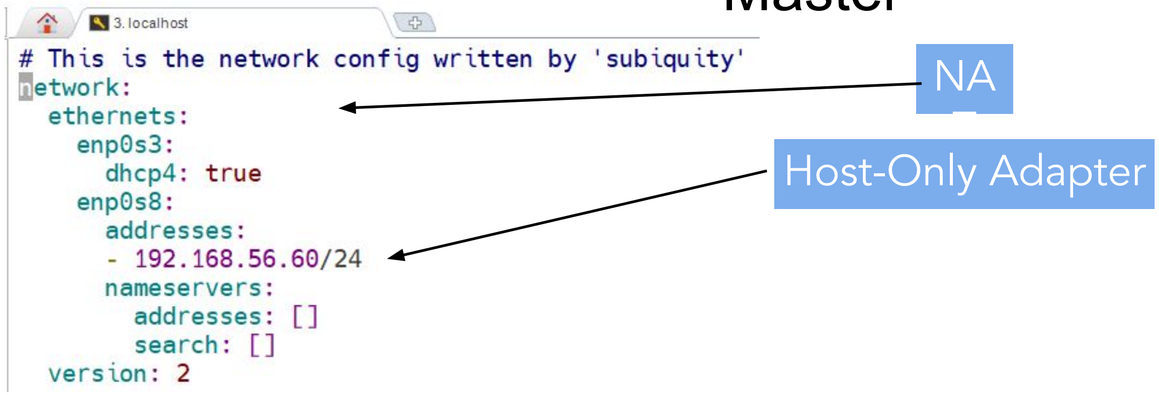
root 비밀번호 설정 및 계정전환

- sudo passwd root
- su - root

Network setup

- check
  vi /etc/netplan/50-cloud-init.yaml

Master



netplan apply

Reference :
https://kubernetes.io/docs/concepts/extend-kubernetes/compute-storage-net/network-plugins/#network-plugin-requirements

Network setup

- vi /etc/sysctl.d/k8s.conf

- 리눅스 시스템에서 커널 파라미터
  sysctl --system

```
net.bridge.bridge-nf-call-iptables  = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward                 = 1
```

net.bridge.bridge-nf-call-iptables = 1

net.bridge.bridge-nf-call-ip6tables = 1

net.ipv4.ip_forward = 1

https://kubernetes.io/docs/setup/production-environment/container-runtimes/#containerd

Network setup

```
cat <<EOF | sudo tee /etc/modules-load.d/k8s.conf
overlay
br_netfilter
EOF

modprobe overlay
modprobe br_netfilter
```

모듈명	역할
`overlay`	컨테이너 이미지 계층 저장 방식 (OverlayFS 사용)
`br_netfilter`	브리지 네트워크의 트래픽이 iptables로 전달되도록 해줌 (쿠버네티스 네트워크 필수)

/etc/modules-load.d/ 디렉터리에 있으므로 systemd가

감지하여 다음 부팅 때 자동으로 모듈을 로딩

명령어	설명
`modprobe overlay`	`overlay` 커널 모듈을 로드하여, OverlayFS (파일시스템 계층화 기술)을 사용 가능하게 만듭니다. 컨테이너 이미지에서 필수입니다.
`modprobe br_netfilter`	`br_netfilter` 모듈을 로드하여, **브리지 네트워크에서 들어오는 트래픽을 iptables로 전달**할 수 있게 만듭니다. 쿠버네티스 네트워킹에 필수입니다.

hostname

vi /etc/cloud/cloud.cfg

```
# This will cause the set+update hostname module to not operate (if true)
preserve_hostname: true
```

쿠버네티스 클러스터의 각 노드들은 고유한 호스트네임을 갖는 것이 중요

호스트네임이 변경되면 클러스터 내에서 노드를 식별하는 데 문제가 발생

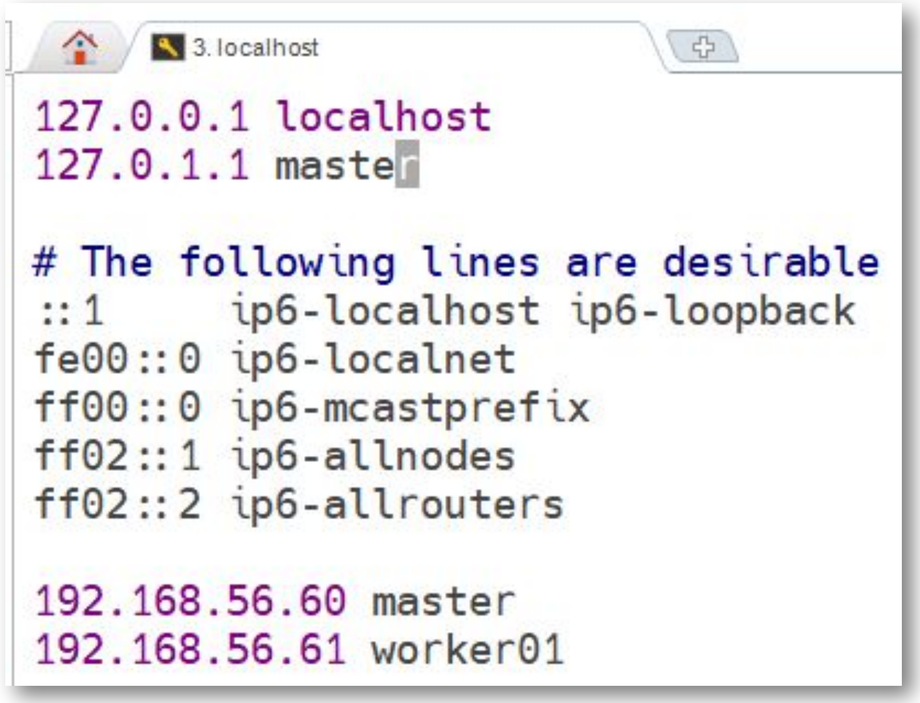
systemctl restart systemd-logind.service

hosts

- vi /etc/hosts

  192.168.56.60 master

  192.168.56.61 worker01



hostname setting

- Hostname setting in master node

```
hostnamectl set-hostname master
```

- If worker’s hostname is equal to master, then you will see following error message on worker node when executing kubeadm join
  >> a Node with name ** and and status "Ready" already exists in the cluster.

Requirements

Docker

- Memory swap off

  swapoff -a

- vi /etc/fstab

```
UUID=a4241556-c44e-4b66-8a52-4a82994c2c99 / ext4 defaults 0 0
#/swap.img      none    swap    sw      0       0
```

```
kopo@master:~$ free -m
               total        used        free
Mem:            2980         236        2083
Swap:              0           0           0
```

재부팅 후 다시 swap이 켜지지 않도록” 막기 위한 조치

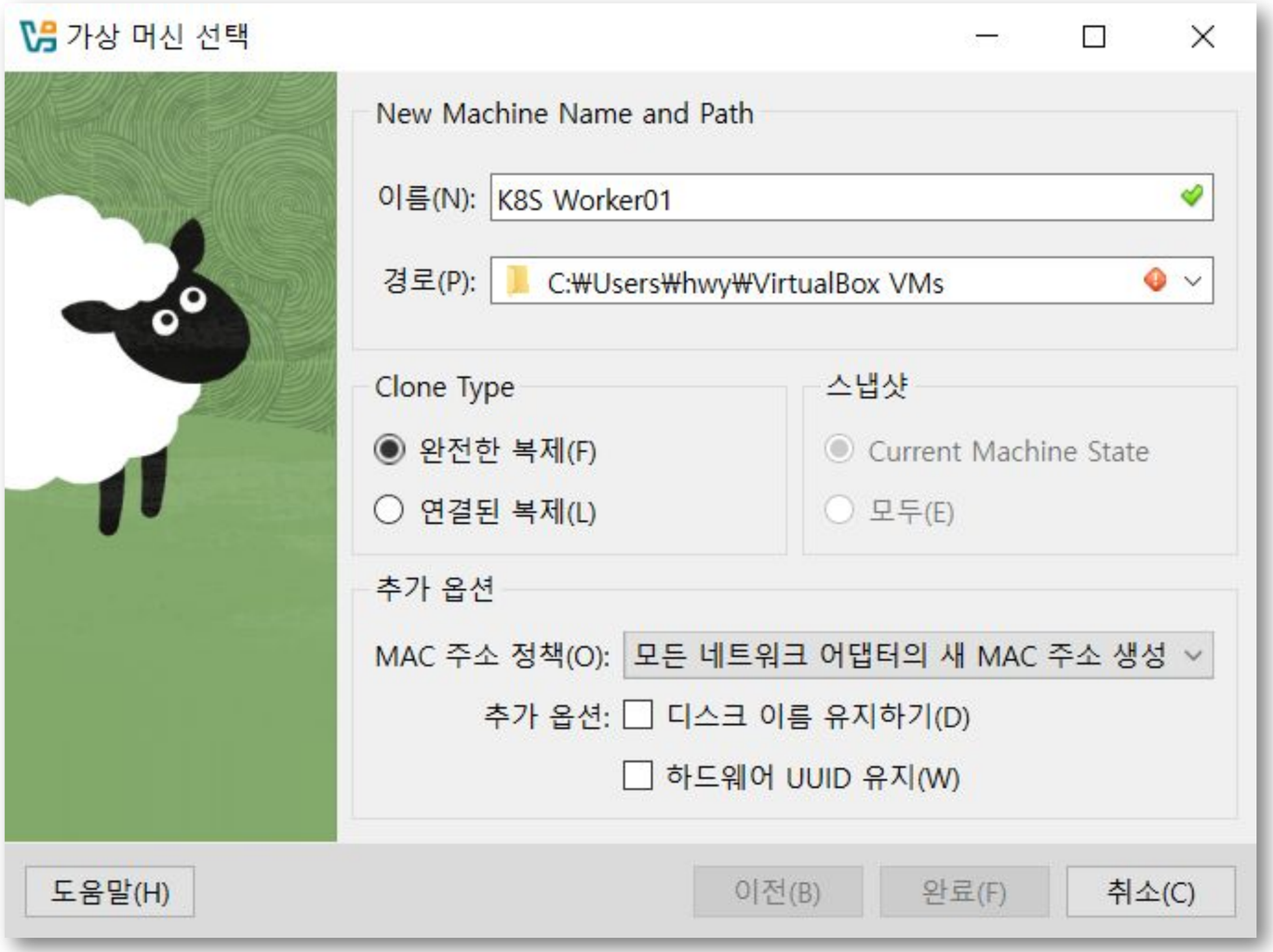
메모리 스왑(Swap)이란?

> **RAM이 부족할 때**, 하드디스크(또는 SSD)의 일부 공간을 **가상 메모리처럼 사용하는 기능**입니다.

운영체제는 실행 중인 프로그램과 데이터를 주기억장치(RAM)에 적재합니다.
하지만 RAM이 부족해지면, 시스템은 **덜 사용되는 메모리 데이터를 디스크로 옮겨서** RAM을 확보합니다.
이때 옮겨지는 공간이 바로 **스왑 영역(swap space)**입니다.

https://kubernetes.io/docs/setup/production-environment/tools/kubeadm/install-kubeadm/

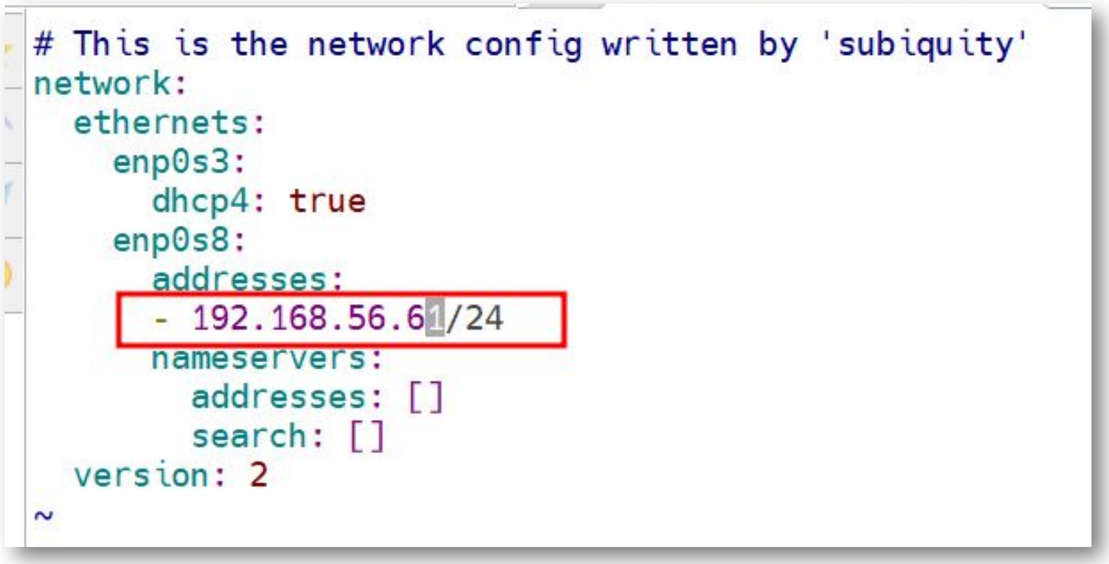
VM 복제


가상 머신 선택
New Machine Name and Path
이름(N): K8S Worker01
경로(P): C:₩Users₩hwy₩VirtualBox VMs
Clone Type
완전한 복제(F)
연결된 복제(L)
스냅샷
Current Machine State
모두(E)
추가 옵션
MAC 주소 정책(O): 모든 네트워크 어댑터의 새 MAC 주소 생성
추가 옵션: 디스크 이름 유지하기(D)
하드웨어 UUID 유지(W)
도움말(H)
이전(B)
완료(F)
취소(C)

VM 복제

worker node

- check
  vi /etc/netplan/50-cloud-init.yaml



netplan
apply

hostname setting

- Hostname setting in worker node

```
hostnamectl set-hostname worker01
```

- If worker’s hostname is equal to master, then you will see following error message on worker node when executing kubeadm join
  >> a Node with name ** and and status "Ready" already exists in the cluster.

Port Open

master(control plane) node

```
ufw allow "OpenSSH"

ufw enable

ufw allow 6443/tcp

ufw allow 2379:2380/tcp

ufw allow 10250/tcp

ufw allow 10259/tcp

ufw allow 10257/tcp

ufw status
```

Worker node

```
ufw allow "OpenSSH"

ufw enable

ufw status

ufw allow 10250/tcp

ufw allow 30000:32767/tcp

ufw status
```

https://kubernetes.io/docs/reference/networking/ports-and-protocols/

Control plane

Protocol	Direction	Port Range	Purpose	Used By
TCP	Inbound	6443	Kubernetes API server	All
TCP	Inbound	2379-2380	etcd server client API	kube-apiserver, etcd
TCP	Inbound	10250	Kubelet API	Self, Control plane
TCP	Inbound	10259	kube-scheduler	Self
TCP	Inbound	10257	kube-controller-manager	Self

Although etcd ports are included in control plane section, you can also host your own etcd cluster externally or on custom ports.

Worker node(s)

Protocol	Direction	Port Range	Purpose	Used By
TCP	Inbound	10250	Kubelet API	Self, Control plane
TCP	Inbound	30000-32767	NodePort Services†	All

[tip] virtual box range port forwarding

1. Vbox register(If needed) : VBoxManage registervm /Users/sf29/VirtualBox\ VMs/Worker/Worker.vbox
2. for i in {30000..30767}; do VBoxManage modifyvm "Worker" --natpf1 "tcp-port$i,tcp,,$i,,$i"; done

Kubernetes 설치

https://kubernetes.io/docs/setup/production-environment/tools/kubeadm/install-kubeadm/

Kubernetes setup

Install containerd

```
apt update
apt install -y containerd
containerd --version
```

```
# containerd 설정 파일 생성
mkdir -p /etc/containerd
containerd config default | sudo tee /etc/containerd/config.toml > /dev/null
# 설정 변경
vi /etc/containerd/config.toml
    SystemdCgroup = true
# containerd 재시작
systemctl restart containerd
# 부팅 시 자동 실행
systemctl enable containerd
# 상태 확인
systemctl is-enabled containerd
systemctl status containerd
```

https://kubernetes.io/docs/setup/production-environment/container-runtimes/#containerd

Kubernetes setup

Install kubernetes

```
apt install apt-transport-https ca-certificates curl -y

# curl -fsSLo /usr/share/keyrings/kubernetes-archive-keyring.gpg https://packages.cloud.google.com/apt/doc/apt-key.gpg

curl https://packages.cloud.google.com/apt/doc/apt-key.gpg | sudo apt-key --keyring /usr/share/keyrings/cloud.google.gpg add -
```

더 이상 유효하지 않음

```
#echo "deb [signed-by=/usr/share/keyrings/kubernetes-archive-keyring.gpg] https://apt.kubernetes.io/ kubernetes-xenial main" | sudo tee /etc/apt/sources.list.d/kubernetes.list

echo "deb [signed-by=/usr/share/keyrings/cloud.google.gpg] https://apt.kubernetes.io/kubernetes-xenial main" | sudo tee /etc/apt/sources.list.d/kubernetes.list

apt update

apt install kubelet kubeadm kubectl

apt-mark hold kubelet kubeadm kubectl
```

#패키지가 자동으로 업그레이드 되지 않도록 고정

기본 도구 및 인증서 설치

구글 클라우드의 GPG 키 추가

쿠버네티스 APT 저장소 추가

패키지 목록 업데이트

쿠버네티스의 주요 구성 요소 설치

설치된 패키지 고정

```
kubeadm version
kubelet --version
kubectl version
```

https://littlemobs.com/blog/kubernetes-package-repository-deprecation/

https://kubernetes.io/ko/docs/setup/production-environment/tools/kubeadm/install-kubeadm/

Kubernetes setup

Install kubernetes

```
apt-get install -y apt-transport-https ca-certificates curl gpg

curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.36/deb/Release.key | sudo gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg

echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.36/deb/ /' | sudo tee /etc/apt/sources.list.d/kubernetes.list


apt-get update

apt-get install -y kubelet kubeadm kubectl

apt-mark hold kubelet kubeadm kubectl
```

기본 도구 및 인증서 설치

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쿠버네티스의 주요 구성 요소 설치

설치된 패키지 고정

```
kubeadm version

kubelet --version

kubectl version
```

https://kubernetes.io/ko/docs/setup/production-environment/tools/kubeadm/install-kubeadm/

CNI(컨테이너 네트워크 인터페이스) 플러그인 설치

```
mkdir -p /opt/bin/

curl -fsSLo /opt/bin/flanneld https://github.com/flannel-io/flannel/releases/download/v0.19.0/flanneld-amd64

chmod +x /opt/bin/flanneld


lsmod | grep br_netfilter

kubeadm config images pull
```

더 이상 유효하지 않음

Master node setting

Kubeadm init

kubeadm init --pod-network-cidr=10.244.0.0/16 \

--apiserver-advertise-address=192.168.56.60 \

--cri-socket=unix:///run/containerd/containerd.sock

Your ip address

```
To start using your cluster, you need to run the following as a regular user:

  mkdir -p $HOME/.kube
  sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
  sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

kubectl 명령을 현재 로그인한 일반 사용자 계정에서 바로 사용

```
Then you can join any number of worker nodes by running the following on each as root:

kubeadm join 192.168.56.60:6443 --token arh6up.usqi6daj82rj4rg2 \
        --discovery-token-ca-cert-hash sha256:12035aced64146fc7ccc5e3e737192c7209bc6bacc3fdb5b14400f6f9fd9adfa
```

worker01에서 수행

```
root@master:/home/kopo# kubectl get nodes
NAME       STATUS     ROLES           AGE     VERSION
master     NotReady   control-plane   7m10s   v1.32.3
worker01   NotReady   <none>          16s     v1.32.3
```

master에서 수행(다음 장 명령 실행 후)

Reference :
https://kubernetes.io/docs/setup/production-environment/tools/kubeadm/create-cluster-kubeadm/

Master node setting

mkdir -p $HOME/.kube

cp -i /etc/kubernetes/admin.conf $HOME/.kube/config

chown $(id -u):$(id -g) $HOME/.kube/config

kubectl cluster-info

kubectl apply -f https://raw.githubusercontent.com/flannel-io/flannel/master/Documentation/kube-flannel.yml

kubectl get pods --all-namespaces

kubectl get nodes -o wide

curl -k https://localhost:6443/version

Reference :
https://kubernetes.io/docs/setup/production-environment/tools/kubeadm/create-cluster-kubeadm/

Worker Node Join

더 이상 유효하지 않음

kubeadm join 192.168.56.60:6443 --token arh6up.usqi6daj82rj4rg2 \

--discovery-token-ca-cert-hash sha256:12035aced64146fc7ccc5e3e737192c7209bc6bacc3fdb5b14400f6f9fd9adfa

master node:

```
root@master:~/.kube# kubectl get pods --all-namespaces
NAMESPACE      NAME                              READY   STATUS    RESTARTS   AGE
kube-flannel   kube-flannel-ds-45m24             1/1     Running   0          56s
kube-flannel   kube-flannel-ds-cjbnw             1/1     Running   0          3m26s
kube-system    coredns-5d78c9869d-55k6l          1/1     Running   0          6m53s
kube-system    coredns-5d78c9869d-ks4sj          1/1     Running   0          6m53s
kube-system    etcd-master                       1/1     Running   0          7m6s
kube-system    kube-apiserver-master             1/1     Running   0          7m8s
kube-system    kube-controller-manager-master    1/1     Running   0          7m6s
kube-system    kube-proxy-6bddb                  1/1     Running   0          56s
kube-system    kube-proxy-hgcdq                  1/1     Running   0          6m53s
kube-system    kube-scheduler-master             1/1     Running   0          7m6s
root@master:~/.kube# kubectl get nodes
NAME       STATUS   ROLES           AGE    VERSION
master     Ready    control-plane   8m     v1.27.2
worker01   Ready    <none>          107s   v1.27.2
```

kubectl get pods --all-namespaces

kubectl get nodes -o wide

curl -k https://localhost:6443/version

Kubectl Autocomplete

k = kubectl

Kubectl autocomplete

BASH

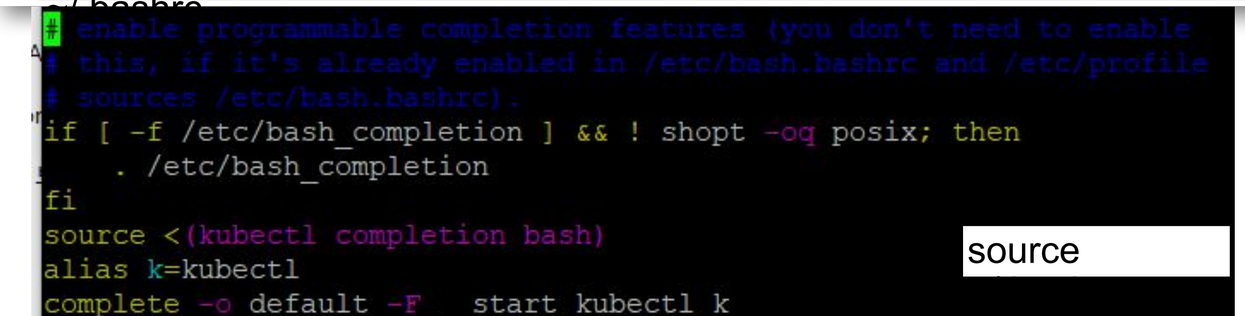
```
source <(kubectl completion bash) # set up autocomplete in bash into the current shell, bash-completion package should be installed first.
echo "source <(kubectl completion bash)" >> ~/.bashrc # add autocomplete permanently to your bash shell.
```

source <(kubectl completion bash) # set up autocomplete in bash into the current shell, bash-completion package should be installed first.

echo "source <(kubectl completion bash)" >> ~/.bashrc # add autocomplete permanently to your bash shell.

You can also use a shorthand alias for `kubectl` that also works with completion:

```
echo "alias k=kubectl" >> ~/.bashrc
echo "complete -o default -F __start_kubectl k" >> ~/.bashrc
```



source

Source : https://kubernetes.io/docs/reference/kubectl/cheatsheet/

bash-completion 로드

```
apt install -y bash-completion
source /usr/share/bash-completion/bash_completion

# ~/.bashrc 하단에 아래와 같이 편집
# bash completion
if [ -f /usr/share/bash-completion/bash_completion ]; then
    source /usr/share/bash-completion/bash_completion
fi

# kubectl completion
source <(kubectl completion bash)

# kubectl alias
alias k=kubectl
complete -o default -F __start_kubectl k
```

Hello world

hello world

- Master node : deploy pod

```
root@kopo:~# kubectl create deployment kubernetes-bootcamp --image=gcr.io/google-samples/kubernetes-bootcamp:v1
deployment.apps/kubernetes-bootcamp created
root@kopo:~# k get deployments.apps
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
kubernetes-bootcamp   0/1     1            0           12s
root@kopo:~# k get deployments.apps
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
kubernetes-bootcamp   1/1     1            1           19s
root@kopo:~# k get pods -o wide
NAME                                  READY   STATUS    RESTARTS   AGE   IP           NODE       NOMINATED NODE   READINESS GATES
kubernetes-bootcamp-6f6656d949-sdhqp   1/1     Running   0          35s   10.244.1.2   worker01   <none>           <none>
```

- Worker node : check the pod is running

```
root@worker01:~# curl http://10.244.1.2:8080
Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-6f6656d949-sdhqp | v=1
```

Flannel Pod 네트워크 통신 오류

Flannel Pod 네트워크 통신 오류

증상

- Worker Node에서는 Pod IP(`10.244.1.2`) 접속 가능
- Control Plane에서는 해당 Pod IP 접속 불가

원인

- VirtualBox VM에 NAT와 Host-Only NIC가 함께 존재
- Flannel이 노드 간 통신용 NIC를 잘못 선택

```
enp0s3 → 10.0.2.15      ❌ NAT, Flannel이 자동 선택
enp0s8 → 192.168.56.x   ✅ Host-Only, 노드 간 통신용
```

로그에서 확인:

```
Using interface with name enp0s3 and address 10.0.2.15
PublicIP: 10.0.2.15
```

kubectl edit daemonset kube-flannel-ds -n kube-flannel

아래 빨간색 부분 추가
args:
- --ip-masq
- --kube-subnet-mgr
- --iface=enp0s8

Terminology and concept

get

kubectl get all

kubectl get nodes

kubectl get nodes -o wide

kubectl get nodes -o yaml

kubectl get nodes -o json

describe

kubectl describe node <node name>

kubectl describe node/<node name>

Other

kubectl exec -it <POD_NAME>

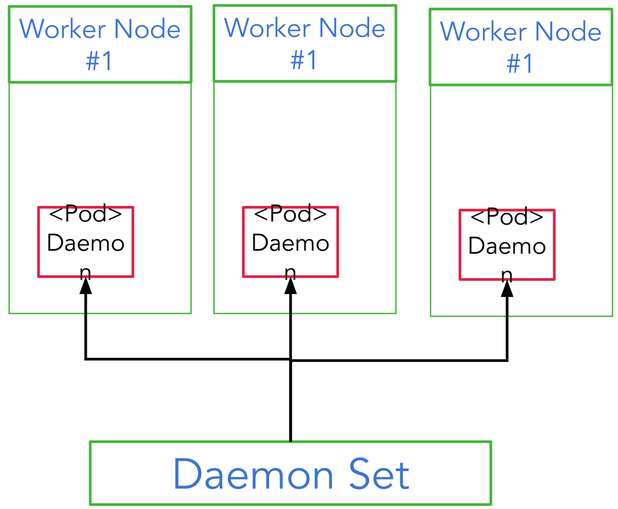
kubectl logs <POD_NAME|TYPE/NAME>

kubectl apply -f <FILENAME>

kubectl delete -f <FILENAME>

Deployment method

- Daemon set : only single container per each node
  예: 로그 수집기, 모니터링 에이전트
- Replica set : control a number of container
  예: 웹 서버 여러 개 실행
- Stateful set : replica set + control sequence
  예: 데이터베이스 클러스터(MySQL, Kafka)



Kubectl practice

https://github.com/wonyongHwang/kopoK8sPractice.git

Make a pod

One container / one pod

- k apply -f first-deploy.yml

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: kopotest
  labels:
    type: app
spec:
  containers:
  - name: app
    image: nginx:latest
```

- k get po
  k get all
  k describe po/kopotest

Liveness/Readiness probe

○ Liveness probe : check after boot up

- 목적: 컨테이너가 살아있는지(동작 중인지) 확인
- 문제 발생 시: 실패하면 컨테이너를 재시작함

○ Readiness probe : check before boot up

- 목적: 컨테이너가 트래픽을 받을 준비가 되었는지 확인
- 문제 발생 시: 실패하면 Service에서 제외시켜 트래픽을 받지 않게 함

```yaml
livenessProbe:
  httpGet:
    path: /health
    port: 8080
  initialDelaySeconds: 10
  periodSeconds: 5

readinessProbe:
  httpGet:
    path: /ready
    port: 8080
  initialDelaySeconds: 5
  periodSeconds: 3
```

◆ `initialDelaySeconds: 10`

- 컨테이너가 **시작된 후, 몇 초 뒤에 처음 probe를 실행할지** 설정
- 예: 컨테이너가 시작되자마자 체크하는 것이 아니라, **10초 후에 처음으로 health check 수행**

◆ `periodSeconds: 5`

- **probe를 주기적으로 수행할 간격(초 단위)**
- 예: 처음 probe가 실행된 후부터 **5초마다** 다음 probe가 실행됨

Make a pod

One container / one pod

k apply -f first-deploy-lp.yml
k apply -f first-deploy-hc.ym

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: kopotest-lp
  labels:
    type: app
spec:
  containers:
  - name: app
    image: nginx:latest
    livenessProbe:
      httpGet:
        path: /not/exists
        port: 80
      initialDelaySeconds: 5
      timeoutSeconds: 2 # Default 1
      periodSeconds: 5 # Defaults 10
      failureThreshold: 1 # Defaults 3
~
```

```yaml
readinessProbe:
  httpGet:
    path: /ready
    port: 8080
  initialDelaySeconds: 5
  periodSeconds: 3
  timeoutSeconds: 2
  failureThreshold: 3
  successThreshold: 2
```

- 컨테이너 시작 후 5초 뒤에 probe 시작
- 3초마다 체크
- 응답이 2초 넘게 걸리면 실패
- 3번 연속 실패해야 실제 실패로 간주
- 2번 연속 성공해야 준비 완료로 간주

```
NAME           READY   STATUS    RESTARTS      AGE   IP           NODE       NOMINATED NODE   READINESS GATES
kopotest-lp    1/1     Running   4 (33s ago)   73s   10.244.1.8   worker01   <none>           <none>
```

Make a pod

Multi container / one pod

- K apply -f multi-container.yml

```
root@kopo:~/project/ex02# k get pod
NAME                                READY   STATUS     RESTARTS   AGE
kubernetes-bootcamp-6f6656d949-sdhqp   1/1     Running    0          10h
two-containers                      1/2     NotReady   0          34m
root@kopo:~/project/ex02# k logs po/two-containers
error: a container name must be specified for pod two-containers, choose one of: [nginx-container
debian-container]
root@kopo:~/project/ex02# k logs po/two-containers nginx-container
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
127.0.0.1 - - [23/Jul/2020:12:52:56 +0000] "GET / HTTP/1.1" 200 42 "-" "curl/7.64.0" "-"
```

```
root@master:~/kopoK8sPractice/Practice/ex02# more multi-container.yml
apiVersion: v1
kind: Pod
metadata:
  name: two-containers
spec:

  restartPolicy: Never

  volumes:
  - name: shared-data
    emptyDir: {}

  containers:

  - name: nginx-container
    image: nginx
    volumeMounts:
    - name: shared-data
      mountPath: /usr/share/nginx/html

  - name: debian-container
    image: debian
    volumeMounts:
    - name: shared-data
      mountPath: /pod-data
    command: ["/bin/sh"]
    args: ["-c", "echo debian 컨테이너에서 안녕하세요 > /pod-data/index.html"]
```

Source : https://kubernetes.io/docs/tasks/access-application-cluster/communicate-containers-same-pod-shared-volume/

Make a pod

Multi container / one pod

You can see that the debian Container has terminated, and the nginx Container is still running.

Get a shell to nginx Container:

```
kubectl exec -it two-containers -c nginx-container -- /bin/bash
```

In your shell, verify that nginx is running:

Do this work-around if dns error occurs…
root@two-containers:/# cat > etc/resolv.conf <<EOF
> nameserver 8.8.8.8
> EOF

```
root@two-containers:/# apt-get update
root@two-containers:/# apt-get install curl procps
root@two-containers:/# ps aux
```

The output is similar to this:

```
USER       PID  ...  STAT START   TIME COMMAND
root         1  ...  Ss   21:12   0:00 nginx: master process nginx -g daemon off;
```

Recall that the debian Container created the `index.html` file in the nginx root directory. Use `curl` to send a GET request to the nginx server:

```
root@two-containers:/# curl localhost
```

The output shows that nginx serves a web page written by the debian container:

```
Hello from the debian container
```

k exec -it two-containers -c nginx-container -- /bin/bash

cd /usr/share/nginx/html/

more index.html

볼륨의 종류 참조 :
https://bcho.tistory.com/1259

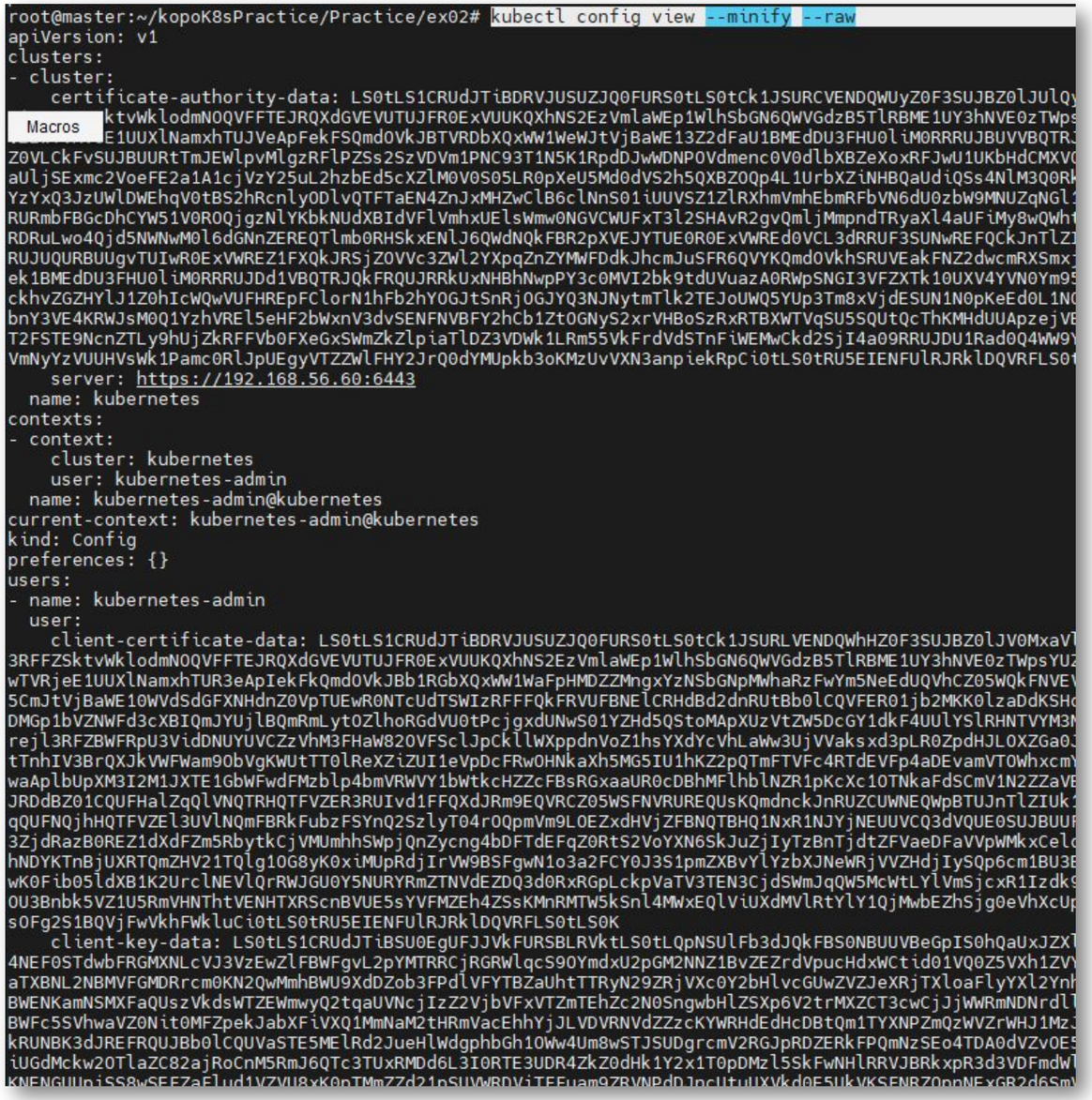
Source : https://kubernetes.io/docs/tasks/access-application-cluster/communicate-containers-same-pod-shared-volume/

LENS 설치

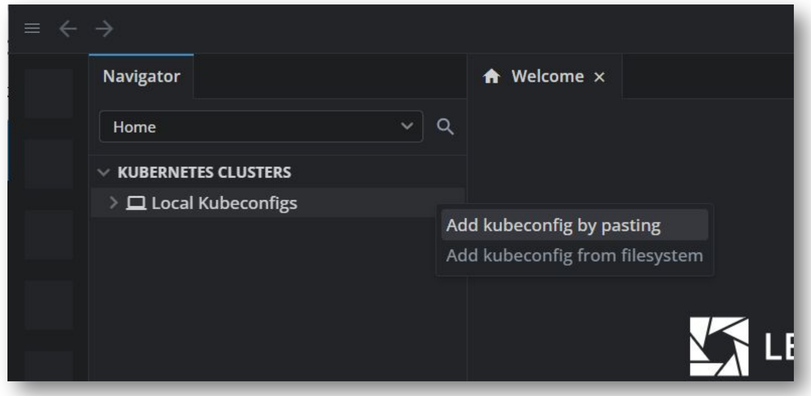
- https://k8slens.dev/

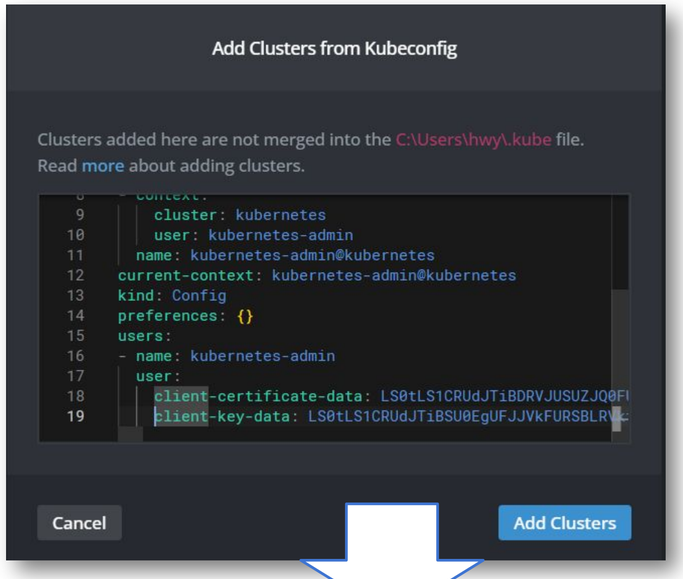
LENS 설치

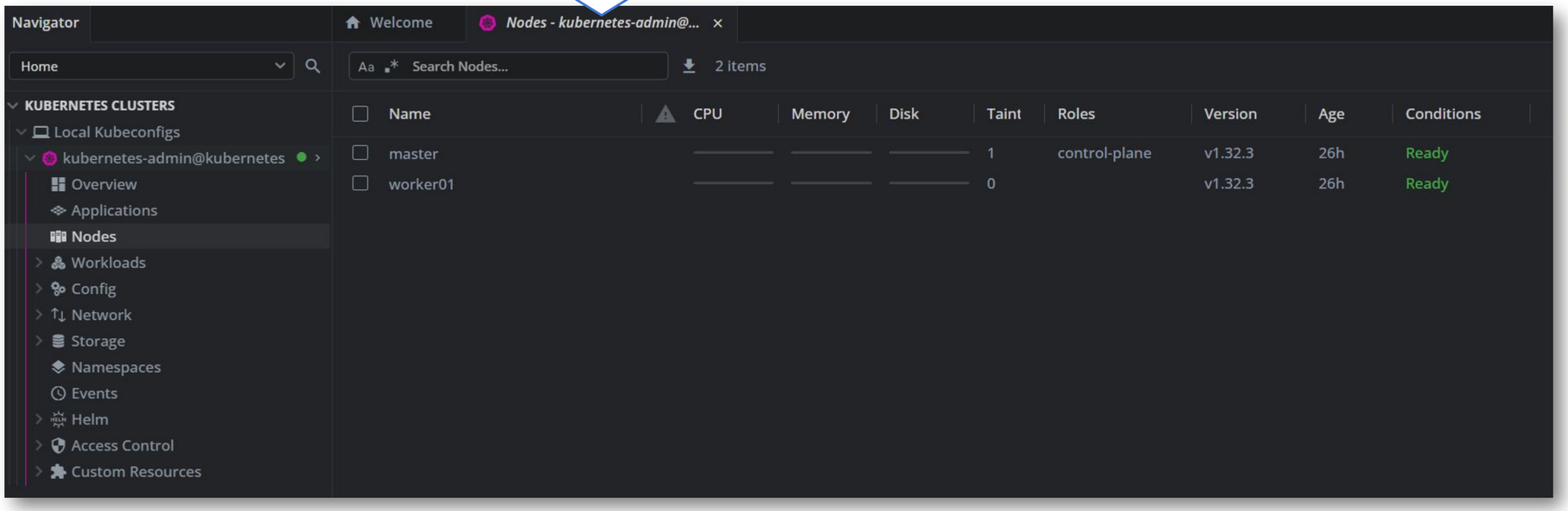
- kubectl config view --minify --raw


root@master:~/kopoK8sPractice/Practice/ex02# kubectl config view --minify --raw
apiVersion: v1
clusters:
- cluster:
certificate-authority-data: LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSURCVENDQWUyZ0F3SUJBZ0lJUlQy
Macros
ktvWklodmNOQVFFTEJRQXdGVEVUTUJFR0ExVUUKQXhNS2EzVmlaWEp1WlhSbGN6QWVGdzB5TlRBME1UY3hNVE0zTWps
E1UUXlNamxhTUJVeApFekFSQmdOVkJBTVRDbXQxWW1WeWJtVjBaWE13Z2dFaU1BMEdDU3FHU0liM0RRRUJBUVVBQTRJ
Z0VLCkFvSUJBUURtTmJEWlpvMlgzRFlPZSs2SzVDVm1PNC93T1N5K1RpdDJwWDNPOVdmenc0V0dlbXBZeXoxRFJwU1UKbHdCMXVC
aUljSExmc2VoeFE2a1A1cjVzY25uL2hzbEd5cXZlM0V0S05LR0pXeU5Md0dVS2h5QXBZOQp4L1UrbXZiNHBQaUdiQSs4NlM3Q0RK
YzYxQ3JzUWlDWEhqV0tBS2hRcnlyODlvQTFTaEN4ZnJxMHZwClB6clNnS01iUUVSZ1ZlRXhmVmhEbmRFbVN6dU0zbW9MNUZqNGl1
RURmbFBGcDhCYW51V0ROQjgzNlYKbkNUdXBIdVFlVmhxUElsWmw0NGVCWUFxT3l2SHAvR2gvQmljMmpndTRyaXl4aUFiMy8wQWh1
RDRuLwo4Qjd5NWNwM0l6dGNnZEREQTlmb0RHSkxENlJ6QWdNQkFBR2pXVEJYTUE0R0ExVWREd0VCL3dRRUF3SUNwREFQCkJnTlZ
RUJUQURBUUgvTUIwR0ExVWREZ1FXQkJRSjZOVVc3ZWl2YXpqZnZYMWFDdkJhcmJuSFR6QVYKQmd0VkhSRUVakFNZ2dwcmRXSmxj
ek1BMEdDU3FHU0liM0RRRUJDd1VBQTRJQkFRQUJRRkUxNHBhNwpPY3c0MVI2bk9tdUVuazA0RWpSNGI3VFZXTk10UXV4YVN0Ym95
ckhvZGZHYlJ1Z0hIcWQwVUFHREpFClorN1hFb2hYOGJtSnRjOGJYQ3NJNytmTlk2TEJoUWQ5YUp3Tm8xVjdESUN1N0pKeEd0L1NC
bnY3VE4KRWJsM0Q1YzhVREl5eHF2bWxnV3dvSENFNVBFY2hCb1ZtOGNyS2xrVHBoSzRxRTBXWTVqSU5SQUtQcThKMHdUUApzejVB
T2FSTE9NcnZTLy9hUjZkRFFVb0FXeGxSWmZkZlpiaTlDZ3VDWk1LRm55VkFrdVdSTnFiWEMwCkd2SjI4a09RRUJDU1Rad0Q4WW9Y
VmNyYzVUUHVsWk1Pamc0RlJpUEgyVTZZWlFHY2JrQ0dYMUpkb3oKMzUvVXN3anpiekRpCi0tLS0tRU5EIENFUlRJRklDQVRFLS0t
server: https://192.168.56.60:6443
name: kubernetes
contexts:
- context:
cluster: kubernetes
user: kubernetes-admin
name: kubernetes-admin@kubernetes
current-context: kubernetes-admin@kubernetes
kind: Config
preferences: {}
users:
- name: kubernetes-admin
user:
client-certificate-data: LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSURLVENDQWhHZ0F3SUJBZ0lJV0MxaVl
3RFFZSktvWklodmNOQVFFTEJRQXdGVEVUTUJFR0ExVUUKQXhNS2EzVmlaWEp1WlhSbGN6QWVGdzB5TlRBME1UY3hNVE0zTWpsYUZ
wTVRjeE1UUXlNamxhTUR3eApIekFkQmdOVkJBb1RGbXQxWW1WaFpHMDZZMngxYzNSbGNpMWhaRzFwYm5NeEdUQVhCZ05WQkFNVEV
5CmJtVjBaWE10WVdSdGFXNHdnZ0VpTUEwR0NTcUdTSWIzRFFFQkFRVUFBNElCRHdBd2dnRUtBb0lCQVFER01jb2MKK0lzaDdKSHc
DMGp1bVZNWFd3cXBIQmJYUjlBQmRmLytOZlhoRGdVU0tPcjgxdUNwS01YZHd5QStoMApXUzVtZW5DcGY1dkF4UUlYSlRHNTVYM3N
rejl3RFZBWFRpU3VidDNUYUVCZzVhM3FHaW82OVFSclJpCkllWXppdnVoZ1hsYXdYcVhLaWw3UjVVaksxd3pLR0ZpdHJLOXZGa0J
tTnhIV3BrQXJkVWFWam9ObVgKWUtTT0lReXZiZUI1eVpDcFRwOHNkaXh5MG5IU1hKZ2pQTmFTVFc4RTdEVFp4aDEvamVTOWhxcmV
waAplbUpXM3I2M1JXTE1GbWFwdFMzblp4bmVRWVY1bWtkcHZZcFBsRGxaaUR0cDBhMFlhblNZR1pKcXc1OTNkaFdSCmV1N2ZZaVB
JRDdBZ01CQUFHalZqQlVNQTRHQTFVZER3RUIvd1FFQXdJRm9EQVRCZ05WSFNVRUREQUsKQmdnckJnRUZCUWNEQWpBTUJnTlZIUk1
qQUFNQjhHQTFVZEl3UVlNQmFBRkFubzFSYnQ2SzlyT04rOQpmVm9LOEZxdHVjZFBNQTBHQ1NxR1NJYjNEUUVCQ3dVQUE0SUJBUUF
3ZjdRazB0REZ1dXdFZm5RbytkCjVMUmhhSWpjQnZycng4bDFTdEFqZ0RtS2VoYXN6SkJuZjIyTzBnTjdtZFVaeDFaVVpWMkxCelc
hNDYKTnBjUXRTQmZHV21TQlg1OG8yK0xiMUpRdjIrVW9BSFgwN1o3a2FCY0J3S1pmZXBvYlYzbXJNeWRjVVZHdjIySQp6cm1BU3B
wK0Fib05ldXB1K2UrclNEVlQrRWJGU0Y5NURYRmZTNVdEZDQ3d0RxRGpLckpVaTV3TEN3CjdSWmJqQW5McWtLYlVmSjcxR1IzdkS
OU3Bnbk5VZ1U5RmVHNThtVENHTXRScnBVUE5sYVFMZEh4ZSsKMnRMTW5kSnl4MWxEQlViUXdMVlRtYlY1QjMwbEZhSjg0eVhXcUp
sOFg2S1BQVjFwVkhFWkluCi0tLS0tRU5EIENFUlRJRklDQVRFLS0tLS0K
client-key-data: LS0tLS1CRUdJTiBSU0EgUFJJVkFURSBLRVktLS0tLQpNSUFb3dJQkFBS0NBUUVBeGpIS0hQaUxJZXl
4NEF0STdwbFRGMXNLcVJ3VzEwZlFBWFgvL2pYMTRRCjRGRWlqcS9OYmdxU2pGM2NNZ1BvZEZrdVpucHdxWCtid01VQ0Z5VXh1ZVY
aTXBNL2NBMVFGMDRrcm0KN2QwMmhBWU9XdDZob3FPdlVFYTBZaUhtTTRyN29ZRjVXc0Y2bHlvcGUwZVZJeXRjTXloaFlyYXl2YnF
BWENKamNSMXFaQUszVkdsWTZEWmwyQ2tqaUVNcjIzZ2VjbVFxVTZmTEhZc2N0SngwbHlZSXp6V2trMXZCT3cwCjJjWWRmNDNrdlI
BWFc5SVhwaVZ0Nit0MFZpekJabXFiVXQ1MmNaM2tHRmVacEhhYjJLVDVRNVdZZzcKYWRHdEdHcDBtQm1TYXNPZmQzWVZrWHJ1MzJ
kRUNBK3dJREFRQUJBb0lCQUVaSTE5MElRd2JueHlWdgphbGh1OWw4Um8wSTJSUDgrcmV2RGJpRDZERkFPQmNzSEo4TDA0dVZvOE5
iUGdMckw2OTlaZC82ajRoCnM5RmJ6QTc3TUxRMDd6L3I0RTE3UDR4ZkZ0dHk1Y2x1T0pDMzl5SkFwNHlRRVJBRkxpR3d3VDFmdWl
KNENGUUpiSS8wSEFZaFlud1VZVU8xK0pIMmZZd21pSUVWRDViTEEuam9ZRVNPdDJpcUtuUXVkd0E5UkVKSENBZ0ppNExGR2d6Sm1

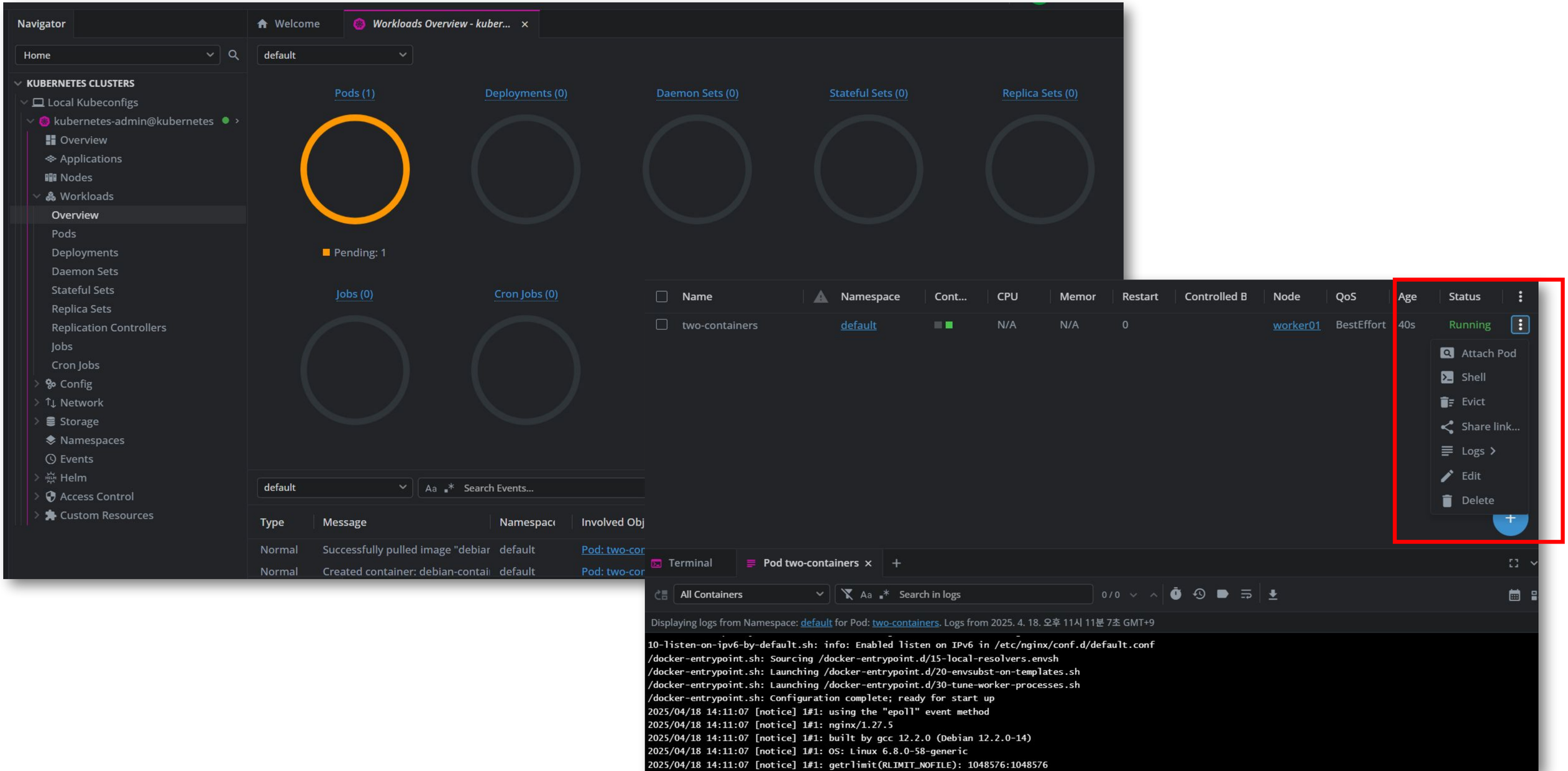
LENS 설치


Navigator
Welcome
Home
KUBERNETES CLUSTERS
Local Kubeconfigs
Add kubeconfig by pasting
Add kubeconfig from filesystem


Add Clusters from Kubeconfig
Clusters added here are not merged into the C:\Users\hwy\.kube file.
Read more about adding clusters.
cluster: kubernetes
user: kubernetes-admin
name: kubernetes-admin@kubernetes
current-context: kubernetes-admin@kubernetes
kind: Config
preferences: {}
users:
- name: kubernetes-admin
user:
client-certificate-data: LS0tLS1CRUdJTiBDRVJUSUZJQ0FU
client-key-data: LS0tLS1CRUdJTiBSU0EgUFJJVkFURSBLRVk
Cancel
Add Clusters

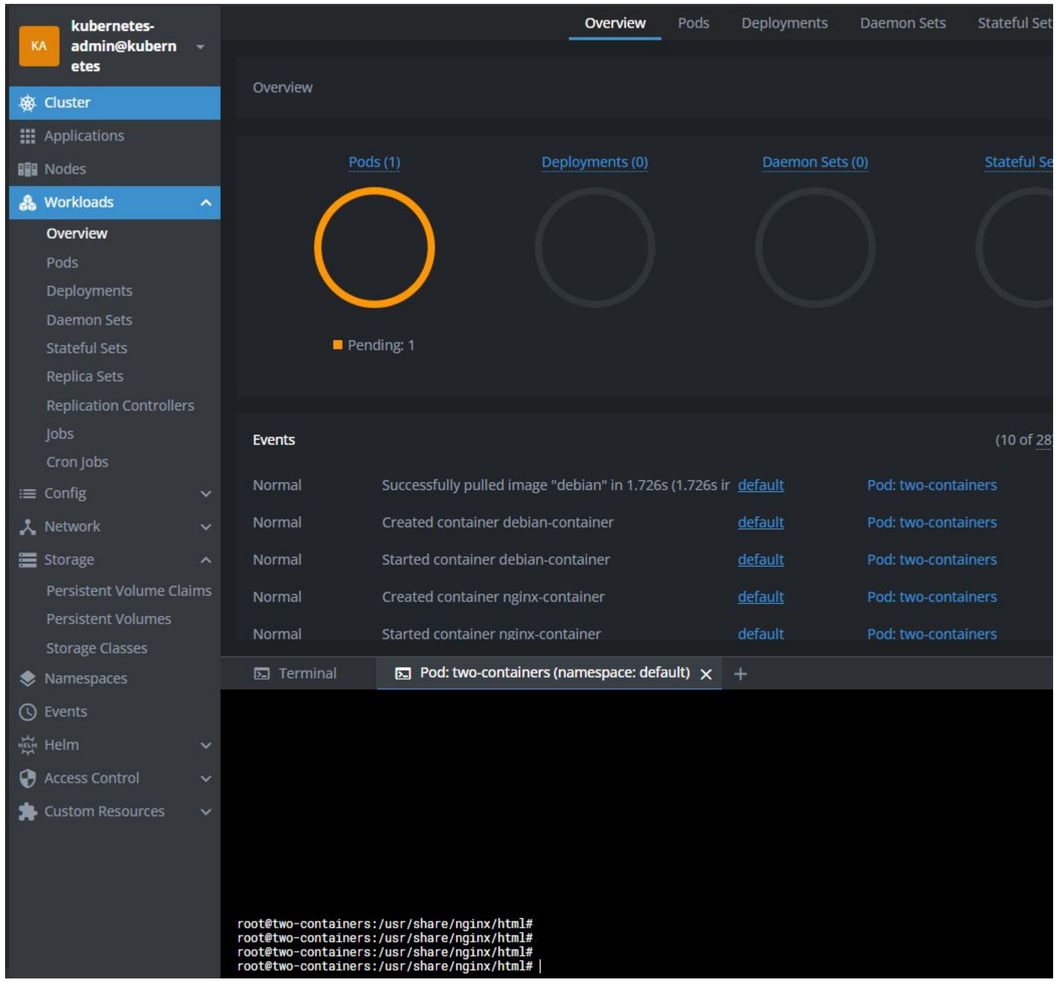

Navigator
Welcome
Nodes - kubernetes-admin@...
Home
Search Nodes...
2 items
KUBERNETES CLUSTERS
Local Kubeconfigs
kubernetes-admin@kubernetes
Overview
Applications
Nodes
Workloads
Config
Network
Storage
Namespaces
Events
Helm
Access Control
Custom Resources
Name
CPU
Memory
Disk
Taint
Roles
Version
Age
Conditions
master
1
control-plane
v1.32.3
26h
Ready
worker01
0
v1.32.3
26h
Ready

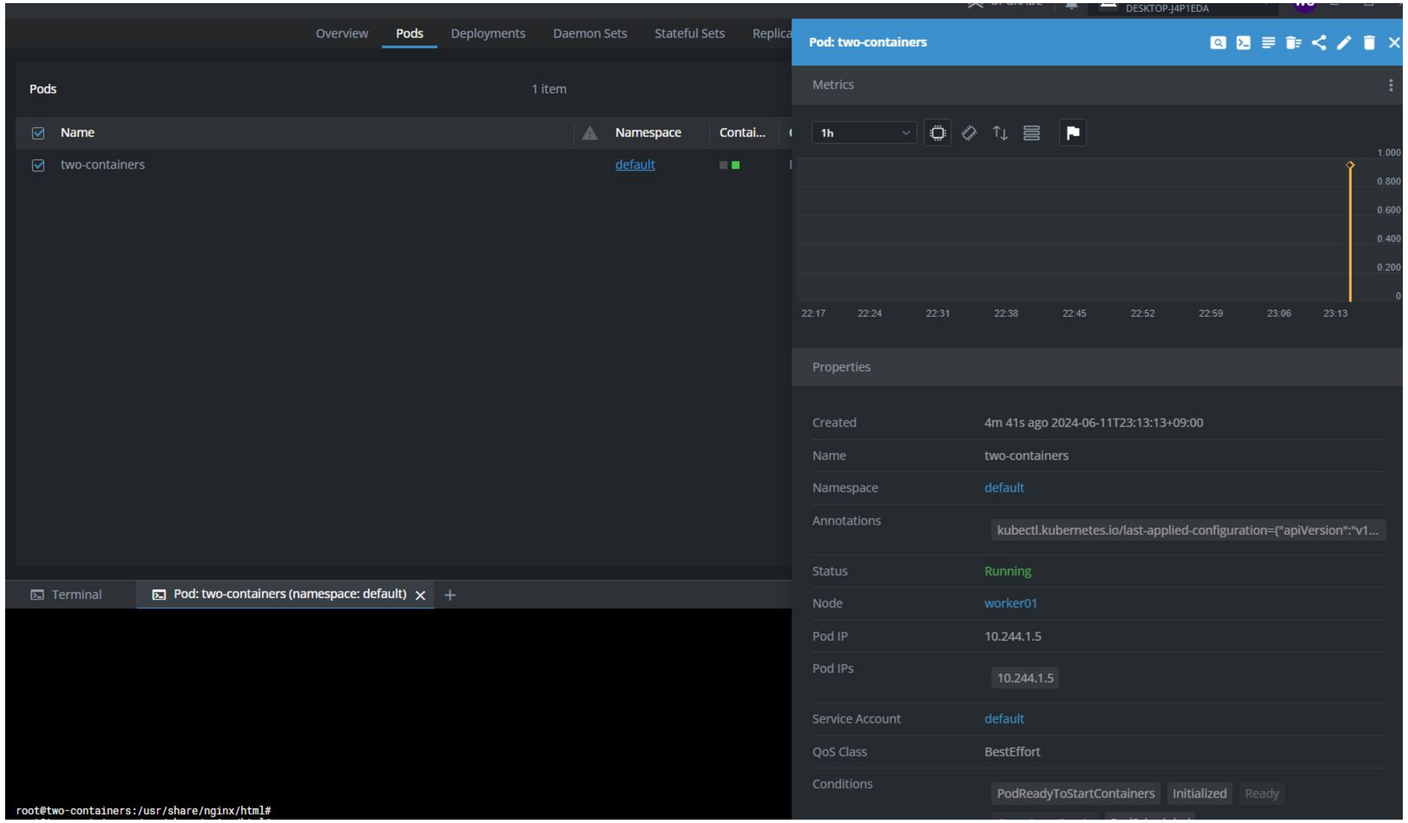
LENS 설치


Navigator
Welcome
Workloads Overview - kuber...
Home
default
KUBERNETES CLUSTERS
Local Kubeconfigs
kubernetes-admin@kubernetes
Overview
Applications
Nodes
Workloads
Overview
Pods
Deployments
Daemon Sets
Stateful Sets
Replica Sets
Replication Controllers
Jobs
Cron Jobs
Config
Network
Storage
Namespaces
Events
Helm
Access Control
Custom Resources
Pods (1)
Deployments (0)
Daemon Sets (0)
Stateful Sets (0)
Replica Sets (0)
Pending: 1
Jobs (0)
Cron Jobs (0)
Search Events...
Type
Message
Namespac
Involved Obj
Normal
Successfully pulled image "debiar
Created container: debian-contai
Name
Namespace
Cont...
CPU
Memor
Restart
Controlled B
Node
QoS
Age
Status
two-containers
N/A
0
worker01
BestEffort
40s
Running
Attach Pod
Shell
Evict
Share link...
Logs
Edit
Delete
Terminal
Pod two-containers
All Containers
Search in logs
Displaying logs from Namespace: default for Pod: two-containers. Logs from 2025. 4. 18. 오후 11시 11분 7초 GMT+9
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/04/18 14:11:07 [notice] 1#1: using the "epoll" event method
2025/04/18 14:11:07 [notice] 1#1: nginx/1.27.5
2025/04/18 14:11:07 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)
2025/04/18 14:11:07 [notice] 1#1: OS: Linux 6.8.0-58-generic
2025/04/18 14:11:07 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576

LENS 설치

- LENS 설정 및 확인





crictl

[참고] 컨테이너 런타임 관리도구

- ctr -n k8s.io container list
- ctr -n k8s.io image list

- crictl
- crictl pods
- crictl images

crictl, ctr은 **현재 노드의** containerd 상태를 확인하는 명령임

Replicas

Replicas

- k apply -f repltest.yml

```yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: frontend
  labels:
    app: guestbook
    tier: frontend
spec:
  # 케이스에 따라 레플리카를 수정한다.
  replicas: 3
  selector:
    matchLabels:
      tier: frontend
  template:
    metadata:
      labels:
        tier: frontend
    spec:
      containers:
      - name: mynignx
        image: nginx:latest
```

- k get rs

selector와 template.labels가 반드시 일치해야 함

이것이 ReplicaSet이 "어떤 Pod를 관리할지"를 결정하는 기준

Source : https://kubernetes.io/ko/docs/concepts/workloads/controllers/replicaset/

Replicas

You can then get the current ReplicaSets deployed:

```
kubectl get rs
```

And see the frontend one you created:

```
NAME       DESIRED   CURRENT   READY   AGE
frontend   3         3         3       6s
```

You can also check on the state of the ReplicaSet:

```
kubectl describe rs/frontend
```

```
root@kopo:~/project/ex03-replica# k get pods --show-labels
NAME                          READY   STATUS    RESTARTS   AGE   LABELS
frontend-jb9kl                1/1     Running   0          54s   tier=frontend
frontend-ksbz8                1/1     Running   0          54s   tier=frontend
frontend-vcpsm                1/1     Running   0          54s   tier=frontend
kubernetes-bootcamp-6f6656d949-sdhqp   1/1     Running   0          10h   app=kubernetes-bootcamp,pod-template-hash=6f6656d949
root@kopo:~/project/ex03-replica# k label pod/frontend-jb9kl tier-
pod/frontend-jb9kl labeled
root@kopo:~/project/ex03-replica# k get pods --show-labels
NAME                          READY   STATUS    RESTARTS   AGE    LABELS
frontend-4jbf6                1/1     Running   0          4s     tier=frontend
frontend-jb9kl                1/1     Running   0          3m6s   <none>
frontend-ksbz8                1/1     Running   0          3m6s   tier=frontend
frontend-vcpsm                1/1     Running   0          3m6s   tier=frontend
kubernetes-bootcamp-6f6656d949-sdhqp   1/1     Running   0          10h    app=kubernetes-bootcamp,pod-template-hash=6f6656d949
root@kopo:~/project/ex03-replica# k label pod/frontend-jb9kl tier=frontend
pod/frontend-jb9kl labeled
root@kopo:~/project/ex03-replica# k get pods --show-labels
NAME                          READY   STATUS    RESTARTS   AGE    LABELS
frontend-jb9kl                1/1     Running   0          3m39s  tier=frontend
frontend-ksbz8                1/1     Running   0          3m39s  tier=frontend
frontend-vcpsm                1/1     Running   0          3m39s  tier=frontend
kubernetes-bootcamp-6f6656d949-sdhqp   1/1     Running   0          10h    app=kubernetes-bootcamp,pod-template-hash=6f6656d949
root@kopo:~/project/ex03-replica# k scale --replicas=6 -f repltest.yml
replicaset.apps/frontend scaled
root@kopo:~/project/ex03-replica# k get pods --show-labels
NAME                          READY   STATUS    RESTARTS   AGE    LABELS
frontend-8vwl6                1/1     Running   0          5s     tier=frontend
frontend-jb9kl                1/1     Running   0          4m20s  tier=frontend
frontend-ksbz8                1/1     Running   0          4m20s  tier=frontend
frontend-lzflt                1/1     Running   0          5s     tier=frontend
frontend-vbthb                1/1     Running   0          5s     tier=frontend
frontend-vcpsm                1/1     Running   0          4m20s  tier=frontend
kubernetes-bootcamp-6f6656d949-sdhqp   1/1     Running   0          10h    app=kubernetes-bootcamp,pod-template-hash=6f6656d949
```

Delete label of one pod

Set the label

Add replica number

Source : https://kubernetes.io/ko/docs/concepts/workloads/controllers/replicaset/

Replicas

k describe rs/<replicas name>

```
root@master:~/kopoK8sPractice/Practice/ex03-replica# k describe rs/frontend
Name:         frontend
Namespace:    default
Selector:     tier=frontend
Labels:       app=guestbook
              tier=afrontend
Annotations:  <none>
Replicas:     3 current / 3 desired
Pods Status:  3 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  tier=frontend
  Containers:
   php-redis:
    Image:        gcr.io/google_samples/gb-frontend:v3
    Port:         <none>
    Host Port:    <none>
    Environment:  <none>
    Mounts:       <none>
  Volumes:        <none>
```

Deployment

Deployment

- k apply -f deploytest.yml

- kubectl get deployments
  kubectl rollout status deployment nginx-deployment
  kubectl get rs
  kubectl get pods --show-labels

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.14.2
        ports:
        - containerPort: 80
```

Source : https://kubernetes.io/docs/concepts/workloads/controllers/deployment/

Deployment

Deployment update

- kubectl edit deployments.apps nginx-deployment

In editor window, modify nginx version
1.14.2 -> 1.16.1

- kubectl rollout status deployment nginx-deployment

```
root@master:~/kopoK8sPractice/Practice/ex04-deployment# k rollout status deployment nginx-deployment
Waiting for deployment "nginx-deployment" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "nginx-deployment" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "nginx-deployment" rollout to finish: 2 out of 3 new replicas have been updated...

Waiting for deployment "nginx-deployment" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "nginx-deployment" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "nginx-deployment" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "nginx-deployment" rollout to finish: 1 old replicas are pending termination...
deployment "nginx-deployment" successfully rolled out
```

- kubectl describe deployments

Source : https://kubernetes.io/docs/concepts/workloads/controllers/deployment/

Deployment

Deployment roll-back

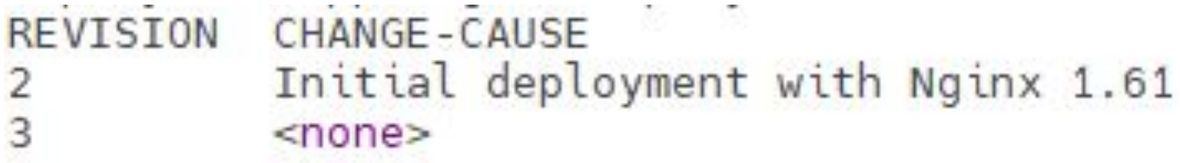
이전 실습 초기화 : kubectl delete -f deploytest.yml

- k apply -f deploytest.yml

k rollout history deployment nginx-deployment
k rollout history deployment nginx-deployment --revision=1

- 버전 업 배포(1.14.2 -> 1.16.1) : k apply -f deploytest2.yml
- kubectl rollout history deployment.v1.apps/nginx-deployment --revision=2
- kubectl rollout undo deployment.v1.apps/nginx-deployment --to-revision=1

Or
kubectl rollout undo deployment.v1.apps/nginx-deployment

- 
```
REVISION  CHANGE-CAUSE
2         Initial deployment with Nginx 1.61
3         <none>
```
ment

Source : https://kubernetes.io/docs/concepts/workloads/controllers/deployment/

Deployment

Scaling

Scaling a Deployment

You can scale a Deployment by using the following command:

```
kubectl scale deployment.v1.apps/nginx-deployment --replicas=10
```

The output is similar to this:

```
deployment.apps/nginx-deployment scaled
```

Assuming horizontal Pod autoscaling is enabled in your cluster, you can setup an autoscaler for your Deployment and choose the minimum and maximum number of Pods you want to run based on the CPU utilization of your existing Pods.

```
kubectl autoscale deployment.v1.apps/nginx-deployment --min=10 --max=15 --cpu-percent=80
```

The output is similar to this:

```
deployment.apps/nginx-deployment scaled
```

Source : https://kubernetes.io/docs/concepts/workloads/controllers/deployment/

deployment

Rolling update

Rolling Update Deployment

The Deployment updates Pods in a rolling update fashion when `.spec.strategy.type==RollingUpdate`. You can specify `maxUnavailable` and `maxSurge` to control the rolling update process.

Max Unavailable

`.spec.strategy.rollingUpdate.maxUnavailable` is an optional field that specifies the maximum number of Pods that can be unavailable during the update process. The value can be an absolute number (for example, 5) or a percentage of desired Pods (for example, 10%). The absolute number is calculated from percentage by rounding down. The value cannot be 0 if `.spec.strategy.rollingUpdate.maxSurge` is 0. The default value is 25%.

For example, when this value is set to 30%, the old ReplicaSet can be scaled down to 70% of desired Pods immediately when the rolling update starts. Once new Pods are ready, old ReplicaSet can be scaled down further, followed by scaling up the new ReplicaSet, ensuring that the total number of Pods available at all times during the update is at least 70% of the desired Pods.

Max Surge

`.spec.strategy.rollingUpdate.maxSurge` is an optional field that specifies the maximum number of Pods that can be created over the desired number of Pods. The value can be an absolute number (for example, 5) or a percentage of desired Pods (for example, 10%). The value cannot be 0 if `MaxUnavailable` is 0. The absolute number is calculated from the percentage by rounding up. The default value is 25%.

For example, when this value is set to 30%, the new ReplicaSet can be scaled up immediately when the rolling update starts, such that the total number of old and new Pods does not exceed 130% of desired Pods. Once old Pods have been killed, the new ReplicaSet can be scaled up further, ensuring that the total number of Pods running at any time during the update is at most 130% of desired Pods.

✅ Rolling Update란?

- **Deployment 리소스**가 사용하는 기본 업데이트 전략
- 기존 버전의 Pod를 **조금씩 줄이면서**, 새로운 버전의 Pod를 **조금씩 늘려서** 배포
- 클러스터 자원과 서비스 가용성을 **최대한 유지**하면서 버전 업그레이드 가능

⚙️ 동작 방식

yaml Copy Edit

```yaml
strategy:
  type: RollingUpdate
  rollingUpdate:
    maxUnavailable: 1
    maxSurge: 1
```

필드	설명
`maxUnavailable`	업데이트 중 동시에 중지될 수 있는 Pod 수 (기본값: 25%)
`maxSurge`	기존 replica 수보다 더 많이 동시에 생성할 수 있는 새 Pod 수 (기본값: 25%)

> 예: replicas: 4, `maxUnavailable: 1`, `maxSurge: 1`
> → 동시에 최대 5개까지 Pod 존재 가능, 최소 3개는 항상 살아있도록 유지

There are many strategies like ‘recreate’, ‘deadline seconds’ and so on.

Source : https://kubernetes.io/docs/concepts/workloads/controllers/deployment/#rolling-update-deployment

참고...

✅ 세 가지 배포 전략 비교표

전략	개요	장점	단점	적용 예
Rolling Update	기존 Pod을 **순차적으로 새 버전으로 교체**	무중단, 자동화 쉬움	구버전과 신버전이 혼재 (문제 발생 시 감지 어려움)	일반 서비스 업데이트
Blue-Green	두 개의 버전(Blue/Green)을 **완전히 분리**하여 배포 후, 트래픽 전환	간단한 롤백, 버전 간 격리	리소스 2배 필요, 배포 자동화 필요	민감한 서비스 업데이트
Canary	소수의 사용자에게만 **새 버전을 점진적으로 배포**하고 반응을 보고 확대	안정성 높음, 실시간 실험 가능	트래픽 라우팅 및 모니터링 복잡	대규모 서비스 실험/검증

Service

Service

cluster ip (internal network)

- k apply -f clusterip.yml

Creating a Service

So we have pods running nginx in a flat, cluster wide, address space. In theory, you could talk to these pods directly,
but what happens when a node dies? The pods die with it, and the **Deployment will create new ones, with different IPs**.
This is the problem a Service solves.
A Kubernetes Service is an abstraction which defines a logical set of Pods running somewhere in your cluster, that all provide the same functionality.
When created, each **Service is assigned a unique IP address (also called clusterIP).**
This address is tied to the lifespan of the Service,
and will not change while the Service is alive.
Pods can be configured to talk to the Service,
and know that communication to the Service will be automatically load-balanced out to some pod that is a member of the Service.

Client → Service (my-nginx:80) → Pod (my-nginx x 2) → nginx 컨테이너

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-nginx
spec:
  selector:
    matchLabels:
      run: my-nginx
  replicas: 2
  template:
    metadata:
      labels:
        run: my-nginx
    spec:
      containers:
      - name: my-nginx
        image: nginx
        ports:
        - containerPort: 80

---
apiVersion: v1
kind: Service
metadata:
  name: my-nginx
  labels:
    run: my-nginx
spec:
  ports:
  - port: 80
    protocol: TCP
  selector:
    run: my-nginx
```

Source : https://kubernetes.io/ko/docs/concepts/services-networking/connect-applications-service/

Service

cluster ip (internal network)

- k get all
- kubectl get pods -l run=my-nginx -o wide
- kubectl get pods -l run=my-nginx -o yaml | grep podIP
- kubectl get svc my-nginx
- kubectl describe svc my-nginx
- kubectl get ep my-nginx
- kubectl exec my-nginx-5dc4865748-rhfq8 -- printenv | grep SERVICE

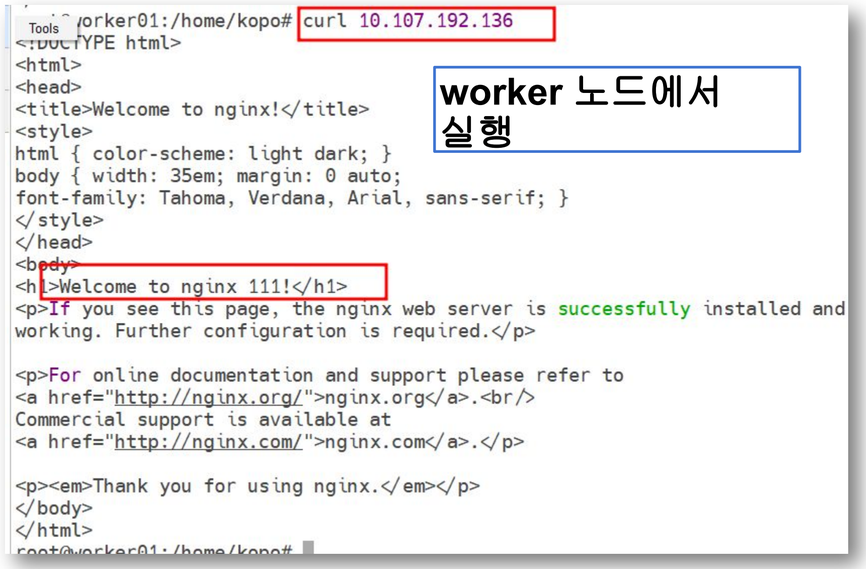
Source : https://kubernetes.io/ko/docs/concepts/services-networking/connect-applications-service/

```
root@kopo:~/project/ex04-deployment# k get all
NAME                                    READY   STATUS    RESTARTS   AGE
pod/kubernetes-bootcamp-6f6656d949-sdhqp   1/1     Running   1          23h
pod/my-nginx-5dc4865748-rhfq8              1/1     Running   0          36s
pod/my-nginx-5dc4865748-z7qkt              1/1     Running   0          36s

NAME                 TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.96.0.1       <none>        443/TCP   2d19h
service/my-nginx     ClusterIP   10.110.148.126  <none>        80/TCP    36s

NAME                                  READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/kubernetes-bootcamp   1/1     1            1           23h
deployment.apps/my-nginx              2/2     2            2           36s

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/kubernetes-bootcamp-6f6656d949   1         1         1       23h
replicaset.apps/my-nginx-5dc4865748              2         2         2       36s
root@kopo:~/project/ex04-deployment# kubectl get pods -l run=my-nginx -o wide
NAME                        READY   STATUS    RESTARTS   AGE   IP           NODE      NOMINATED NODE
READINESS GATES
my-nginx-5dc4865748-rhfq8   1/1     Running   0          60s   10.244.1.28   worker01   <none>           <none>
my-nginx-5dc4865748-z7qkt   1/1     Running   0          60s   10.244.1.27   worker01   <none>           <none>
root@kopo:~/project/ex04-deployment# kubectl get pods -l run=my-nginx -o yaml | grep podIP
        f:podIP: {}
        f:podIPs:
    podIP: 10.244.1.28
    podIPs:
        f:podIP: {}
        f:podIPs:
    podIP: 10.244.1.27
    podIPs:
root@kopo:~/project/ex04-deployment# kubectl get svc my-nginx
NAME       TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
my-nginx   ClusterIP   10.110.148.126  <none>        80/TCP    104s
root@kopo:~/project/ex04-deployment# kubectl describe svc my-nginx
Name:              my-nginx
Namespace:         default
Labels:            run=my-nginx
Annotations:       Selector:  run=my-nginx
Type:              ClusterIP
IP:                10.110.148.126
Port:              <unset>  80/TCP
TargetPort:        80/TCP
Endpoints:         10.244.1.27:80,10.244.1.28:80
Session Affinity:  None
Events:            <none>
root@kopo:~/project/ex04-deployment# kubectl get ep my-nginx
NAME       ENDPOINTS                       AGE
my-nginx   10.244.1.27:80,10.244.1.28:80   2m3s

root@kopo:~/project/ex04-deployment# kubectl exec my-nginx-5dc4865748-rhfq8 -- printenv | grep
SERVICE
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_SERVICE_HOST=10.96.0.1
MY_NGINX_SERVICE_PORT=80
KUBERNETES_SERVICE_PORT=443
```

클러스터 IP로 접속 확인

Pod <-> Local File Copy

- Pod의 Nginx 에서 서비스하는 index.html 파일을 로컬로 복사
  k cp default/my-nginx-646554d7fd-gqsfh:usr/share/nginx/html/index.html ./index.html
- index.html을 편집하여 Pod간 index.html 파일 내용을 다르게 구분해 둔다.
- 편집한 index.html 파일을 Pod로 복사
  k cp ./index.html default/my-nginx-646554d7fd-gqsfh:usr/share/nginx/html/index.html
- 해당 Pod로 진입하여 변경된 파일 확인
  k exec -it my-nginx-646554d7fd-gqsfh -- /bin/bash
- 접속 확인

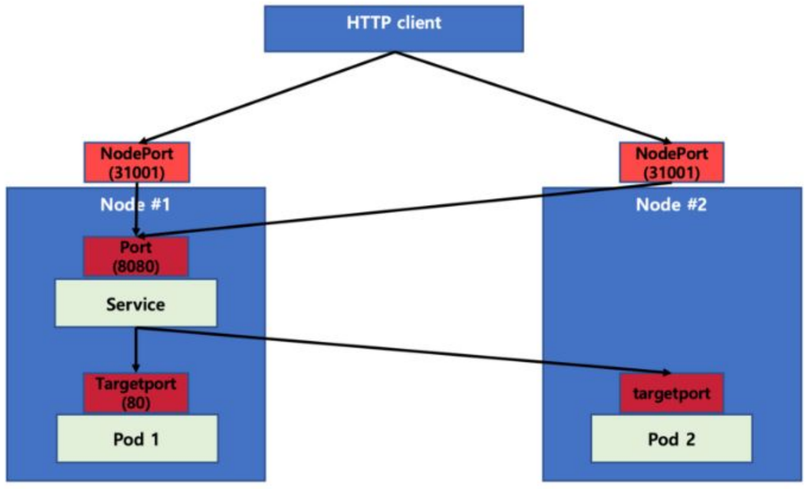


Service

Nodeport (expose network)

상황	설명
🧪 개발/테스트 환경에서 외부 브라우저로 직접 접속하고 싶을 때	예: curl http://<node-ip>:30080 또는 브라우저에서 Nginx 접속
🌐 클라우드가 아닌 Bare Metal 환경에서 외부 노출이 필요할 때	LoadBalancer가 없기 때문에 NodePort가 유일한 대안
🔧 Ingress를 구성하기 전에 임시로 노출해야 할 때	프론트엔드/백엔드 서비스 검증용 등

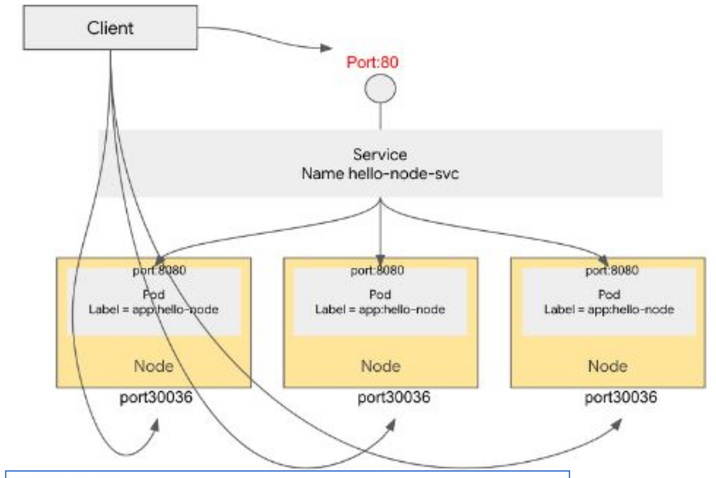
```
spec:
  ports:
  - nodePort: 31001
    port: 8080
    targetPort: 80
    protocol: TCP
  selector:
    app: helloworld
  type: NodePort
```



그림출처:
https://m.blog.naver.com/PostView.naver?isHttpsRedirect=true&blogId=freepsw&logNo=221910012471

```
type: NodePort
ports:
  - name: http
    port: 80
    protocol: TCP
    targetPort: 8080
    nodePort: 30036
```

아래 그림과 같은 구조가 된다.



그림출처:
https://bcho.tistory.com/tag/nodeport

Source :

Service

Nodeport (expose network)

- K apply -f nodeport.yml

curl http://192.168.56.61:30101

Name	my-nginx
Namespace	default
Labels	run=my-nginx
Annotations	kubectl.kubernetes.io/last-applie
Selector	run=my-nginx
Type	NodePort
Session Affinity	None

Connection

Cluster IP	10.109.174.238
Cluster IPs	10.109.174.238
IP families	IPv4
IP family policy	SingleStack
Ports	80:30101/TCP

Endpoint Slices

Name	Endpoints	Ports
my-nginx-n5lms	2/2	80/TCP

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-nginx
spec:
  selector:
    matchLabels:
      run: my-nginx
  replicas: 2
  template:
    metadata:
      labels:
        run: my-nginx
    spec:
      containers:
      - name: my-nginx
        image: nginx
        ports:
        - containerPort: 80

---
apiVersion: v1
kind: Service
metadata:
  name: my-nginx
  labels:
    run: my-nginx
spec:
  type: NodePort
  ports:
  - protocol: TCP
    nodePort: 30101
    port: 80
    targetPort: 80
  selector:
    run: my-nginx
```

Reference: https://bcho.tistory.com/tag/nodeport

Service

at worker node

```
root@master:~/kopoK8sPractice/Practice/ex06-nodeport# k describe svc my-nginx
Name:                     my-nginx
Namespace:                default
Labels:                   run=my-nginx
Annotations:              <none>
Selector:                 run=my-nginx
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.108.110.100
IPs:                      10.108.110.100
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
NodePort:                 <unset>  30101/TCP
Endpoints:                10.244.1.65:80,10.244.1.64:80
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>
```

curl 10.108.110.100

curl localhost:30101

curl 10.244.1.64:80

curl 10.244.1.65:80

정리

타입	접근 가능 위치	기본 포트	사용 목적
ClusterIP (기본값)	클러스터 내부에서만 접근 가능	내부 IP	마이크로서비스 간 통신, 백엔드 호출
NodePort	클러스터 **외부에서 접근 가능**	노드의 IP + 고정 포트 (30000~32767)	브라우저/사용자 직접 접근, 외부 테스트
LoadBalancer	클라우드 환경에서 퍼블릭 IP 제공	외부 IP	운영 서비스에 외부 트래픽 수용

LoadBalancer

개요

- Pod들과 통신하기 위한 **Service 리소스** 생성

```yaml
spec:
  type: LoadBalancer
```

- **외부에 노출되는 퍼블릭 IP를 제공**하는 서비스 타입
- 클라우드 환경(GKE, EKS, AKS 등)에서 외부 IP를 자동으로 할당
- **Bare Metal 환경에서는 MetalLB 같은 외부 LoadBalancer가 필요**


```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-nginx
spec:
  selector:
    matchLabels:
      run: my-nginx
  replicas: 2
  template:
    metadata:
      labels:
        run: my-nginx
    spec:
      containers:
      - name: my-nginx
        image: nginx
        ports:
        - containerPort: 80

---
apiVersion: v1
kind: Service
metadata:
  name: my-nginx
  labels:
    run: my-nginx
spec:
  type: LoadBalancer
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  selector:
    run: my-nginx
```

개요

✅ 실행 시 기대 결과

1. 클라우드 환경이면 `kubectl get svc` 시 외부 IP가 할당됨

bash Copy Edit

```
kubectl get svc my-nginx
```

pgsql Copy Edit

```
NAME        TYPE           CLUSTER-IP     EXTERNAL-IP     PORT(S)        AGE
my-nginx    LoadBalancer   10.96.0.123    34.80.12.45     80:32456/TCP   2m
```

2. 브라우저나 curl로 접속:

nginx Copy Edit

```
curl http://<EXTERNAL-IP>
```

MetalLB

로컬/Bare Metal 환경에 클라우드처럼 외부 IP를 할당할 수 있게 해주는 네트워크 로드밸런서

- 클라우드 환경(GKE, EKS 등)은 LoadBalancer 타입 사용 시 외부 IP 자동 할당됨
- Bare Metal 환경에서는 `LoadBalancer` Service에 외부 IP를 할당·광고해 줄 구현체가 기본 제공되지 않음

```
kubectl apply -f metallb-native.yaml
```

```
kubectl get pods -n metallb-system
```

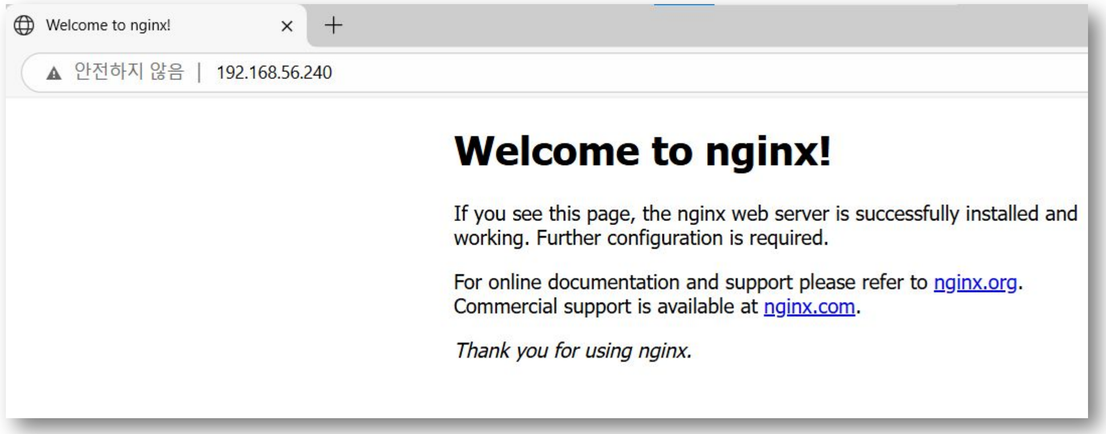
```
kubectl apply -f metallb-config.yaml
```

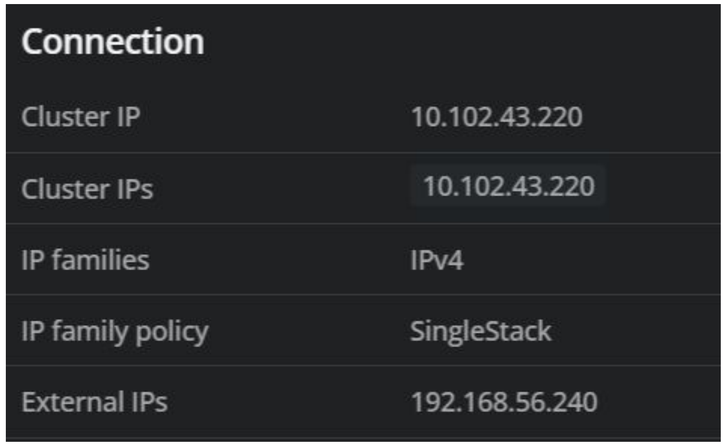
```
kubectl get ipaddresspool,l2advertisement -n metallb-system
```

결과 확인

k apply -f loadbalancer.yml

```
# kubectl get svc
NAME         TYPE           CLUSTER-IP      EXTERNAL-IP      PORT(S)        AGE
kubernetes   ClusterIP      10.96.0.1       <none>           443/TCP        2d1h
my-nginx     LoadBalancer   10.111.185.146  192.168.56.240   80:31298/TCP   27m
```






```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-nginx
spec:
  selector:
    matchLabels:
      run: my-nginx
  replicas: 2
  template:
    metadata:
      labels:
        run: my-nginx
    spec:
      containers:
      - name: my-nginx
        image: nginx
        ports:
        - containerPort: 80

---
apiVersion: v1
kind: Service
metadata:
  name: my-nginx
  labels:
    run: my-nginx
spec:
  type: LoadBalancer
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  selector:
    run: my-nginx
```

Ingress

클러스터 외부의 HTTP/HTTPS 요청을 클러스터 내부의 서비스(Service)로 라우팅 해주는 리소스

여러 도메인과 경로를 하나의 엔드포인트로 받아 처리할 수 있도록 해주는 **트래픽 라우터**

Ingress

https://kubernetes.github.io/ingress-nginx/deploy/#bare-metal

- wget https://raw.githubusercontent.com/kubernetes/ingress-nginx/controller-v1.15.1/deploy/static/provider/baremetal/deploy.yaml
- k apply -f deploy.yaml

kubectl get pods -n ingress-nginx

kubectl get svc -n ingress-nginx

kubectl get ingressclass

kubectl get all -n ingress-nginx

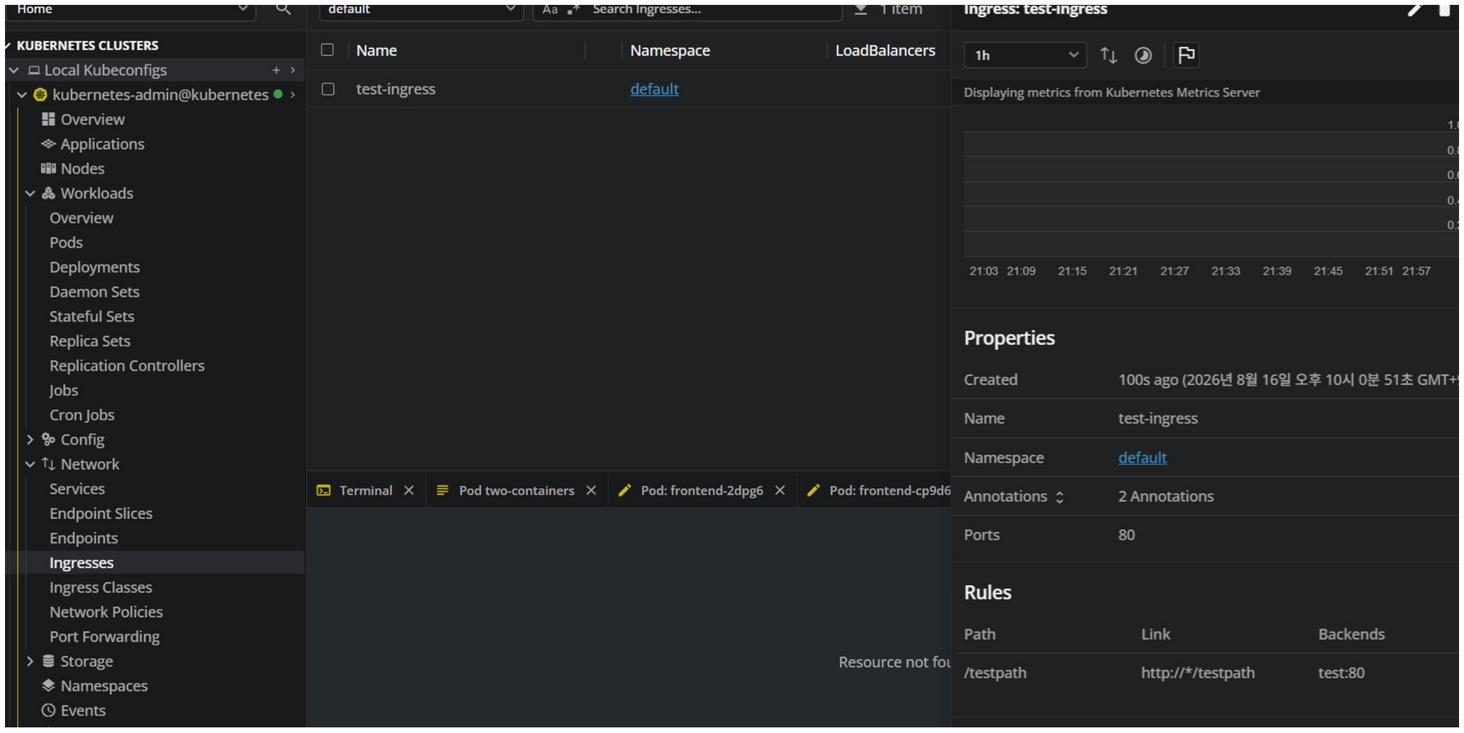
ingress test

basic.yml

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: test-ingress
  annotations:

nginx.ingress.kubernetes.io/rewrite-target: /
spec:
  rules:
  - http:
      paths:
      - path: /testpath
        pathType: Prefix
        backend:
          service:
            name: test
```

root@kopo:~/project/ex08-ingress**# k get ingress**

```
root@master:/home/kopo/kopoK8sPractice/Practice/ex08-ingress# k get ingress
NAME           CLASS    HOSTS   ADDRESS   PORTS   AGE
test-ingress   <none>   *                 80      61s
```



Ingress

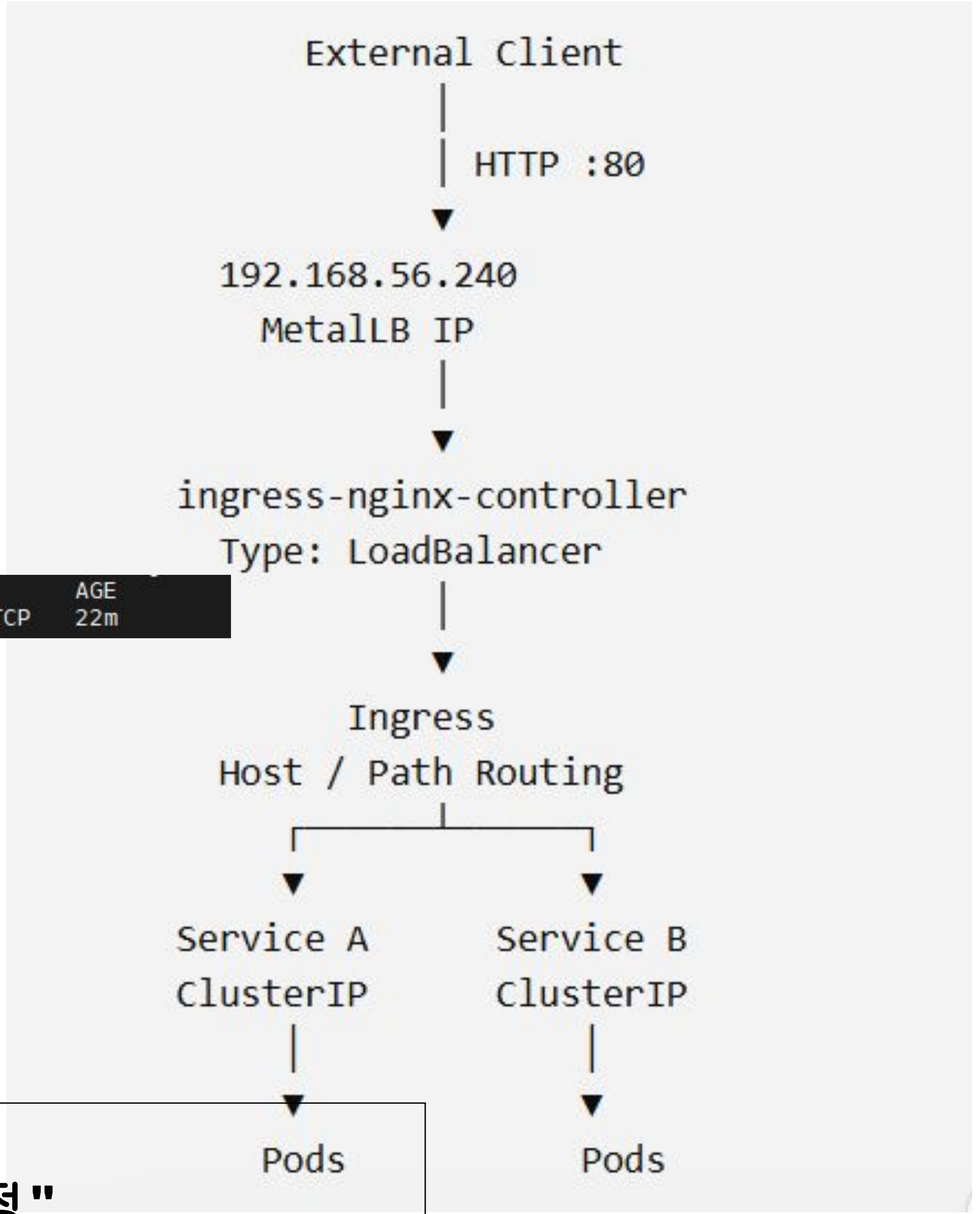
Ingress Controller 앞에 `LoadBalancer` Service를 두어 외부 IP를 확보하는 구성

```
kubectl patch svc ingress-nginx-controller \
  -n ingress-nginx \
  -p '{"spec":{"type":"LoadBalancer"}}'
```

확인>

```
kubectl get svc ingress-nginx-controller -n ingress-nginx
```

```
NAME                       TYPE           CLUSTER-IP     EXTERNAL-IP      PORT(S)                      AGE
ingress-nginx-controller   LoadBalancer   10.105.44.81   192.168.56.240   80:32465/TCP,443:30204/TCP   22m
```



MetalLB = "외부에서 들어올 IP를 제공"
Ingress = "들어온 HTTP 요청을 어느 Service로 보낼지 결정"

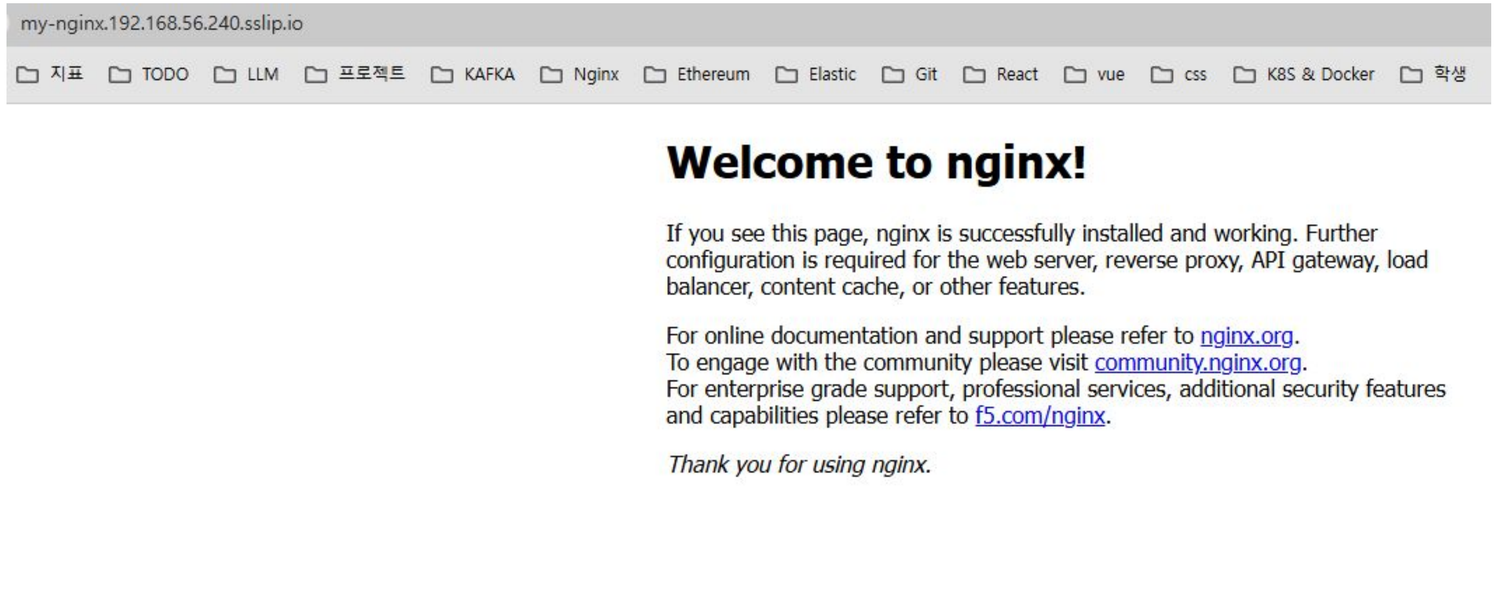
Ingress

Apply and test

- k apply -f ingress.yml

```yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: my-nginx
  annotations:
    ingress.kubernetes.io/rewrite-target: "/"
    ingress.kubernetes.io/ssl-redirect: "false"
spec:
  ingressClassName: "nginx"
  rules:
    - host: my-nginx.192.168.56.240.sslip.io
      http:
        paths:
        - path: /
          pathType: Prefix
          backend:
            service:
              name: my-nginx
              port:
                number: 80

---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-nginx
spec:
  selector:
    matchLabels:
      run: my-nginx
  replicas: 2
  template:
    metadata:
      labels:
        run: my-nginx
    spec:
      containers:
      - name: my-nginx
        image: nginx
        ports:
        - containerPort: 80

---
apiVersion: v1
kind: Service
metadata:
  name: my-nginx
  labels:
    run: my-nginx
spec:
  type: ClusterIP
```

```
kubectl get pods -n ingress-nginx
kubectl get svc -n ingress-nginx
kubectl get ingressclass
kubectl get all -n ingress-nginx
```



Practice #1

이전의 Docker 실습에서 ...

- docker login
- docker tag springboot-mysql-openjdk21-springboot-app:latest wyhwang/spboot_app_jdk21:latest
- docker push wyhwang/spboot_app_jdk21:latest

실습

mysql-pvc가 mysql-pv 실습을 위해서 /mnt/data/mysql 경로는 테스트 환경에서 존재해야 하며, 필요 시 수동 생성:

sudo mkdir -p /mnt/data/mysql sudo chmod 777 /mnt/data/mysql

springboot-deployment.yaml에서 docker image 경로 지정 확인(docker hub id/image name:version)

```
containers:
     - name: springboot-app
       image: wyhwang/spboot_app_jdk21:latest  # 빌드 후 Docker Hub에 올려야 함
       ports:
       - containerPort: 8080
```

실행 순서

kubectl apply -f mysql-volume.yaml

kubectl apply -f mysql-deployment.yaml

kubectl apply -f mysql-service.yaml

kubectl apply -f springboot-deployment.yaml

kubectl apply -f springboot-service.yaml

결과 확인

확인

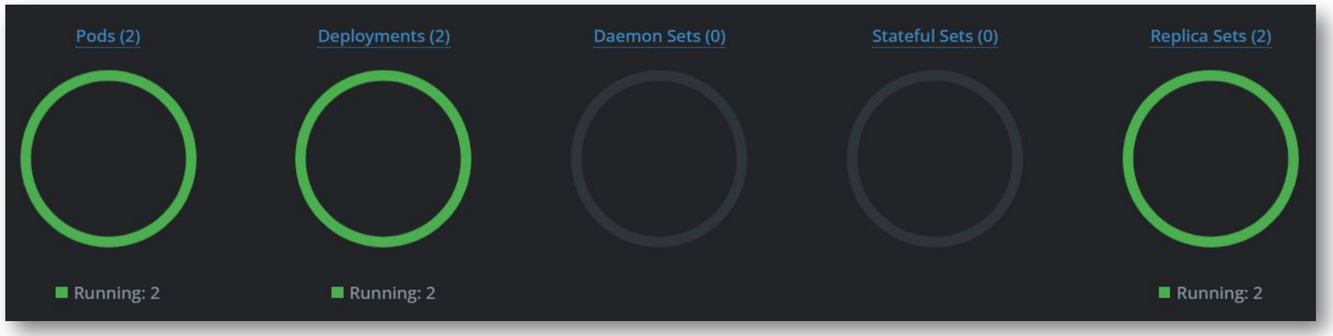
kubectl get svc springboot-app

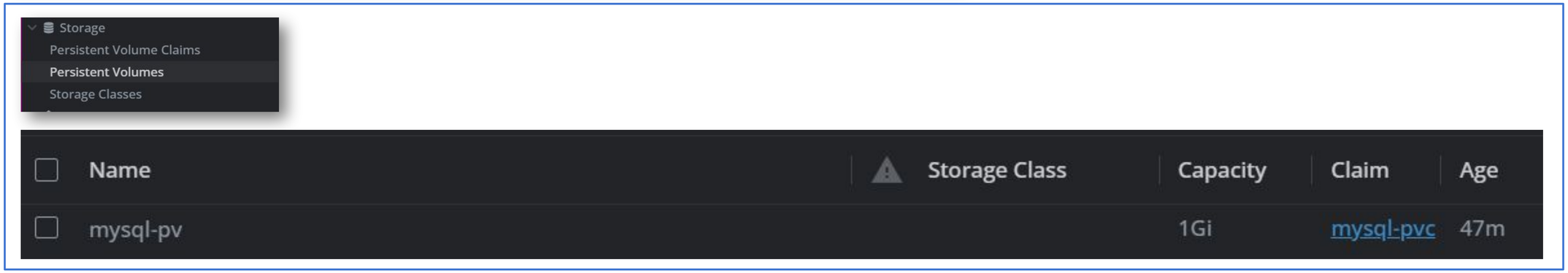
```
NAME             TYPE       CLUSTER-IP     EXTERNAL-IP   PORT(S)          AGE
springboot-app   NodePort   10.107.99.89   <none>        8087:30810/TCP   2m52s
```

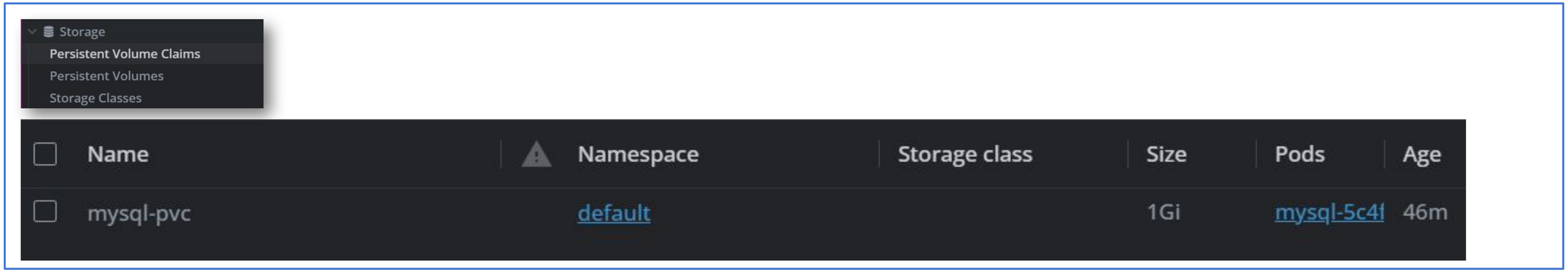
curl http://192.168.56.61:30810

```
Hello from Spring Boot with OpenJDK 21 and MySQL!root@master:~/kopoDockerPractice/springboot-mysql-openjdk21/kubernetes
```

결과 확인


Pods (2)
Deployments (2)
Daemon Sets (0)
Stateful Sets (0)
Replica Sets (2)
Running: 2
Running: 2
Running: 2


Storage
Persistent Volume Claims
Persistent Volumes
Storage Classes
Name
Storage Class
Capacity
Claim
Age
mysql-pv
1Gi
mysql-pvc
47m


Storage
Persistent Volume Claims
Persistent Volumes
Storage Classes
Name
Namespace
Storage class
Size
Pods
Age
mysql-pvc
default
1Gi
mysql-5c4f
46m

기타

[compose] Docker Compose를 Kubernetes YAML로 바꿔주는 자동화 도구

🔧 Kompose란?

- **공식 사이트:** https://kompose.io
- **GitHub:** https://github.com/kubernetes/kompose

> **주요 목적:**
> Docker Compose로 작성된 프로젝트를 빠르게 Kubernetes 환경으로 마이그레이션하기 위함입니다.

✅ Kompose 주요 기능

기능	설명
`docker-compose.yml` → `*.yaml`	Kubernetes Deployment, Service, PVC 등으로 자동 변환
네트워크, 환경변수, 볼륨 등 처리	Compose의 설정을 Kubernetes 형식에 맞게 변환
다양한 드라이버 지원	OpenShift, Helm 등으로도 변환 가능 (옵션)

🚀 사용 방법

1. `docker-compose.yml`이 있는 디렉토리로 이동
2. 다음 명령어 실행:

```bash
kompose convert
```

3. 결과:

```bash
mysql-service.yaml
mysql-deployment.yaml
springboot-app-deployment.yaml
springboot-app-service.yaml
```

4. 적용:

```bash
kubectl apply -f .
```

```
curl -L https://github.com/kubernetes/kompose/releases/download/v1.30.0/kompose-linux-amd64 -o kompose

chmod +x kompose

sudo mv kompose /usr/local/bin/
```

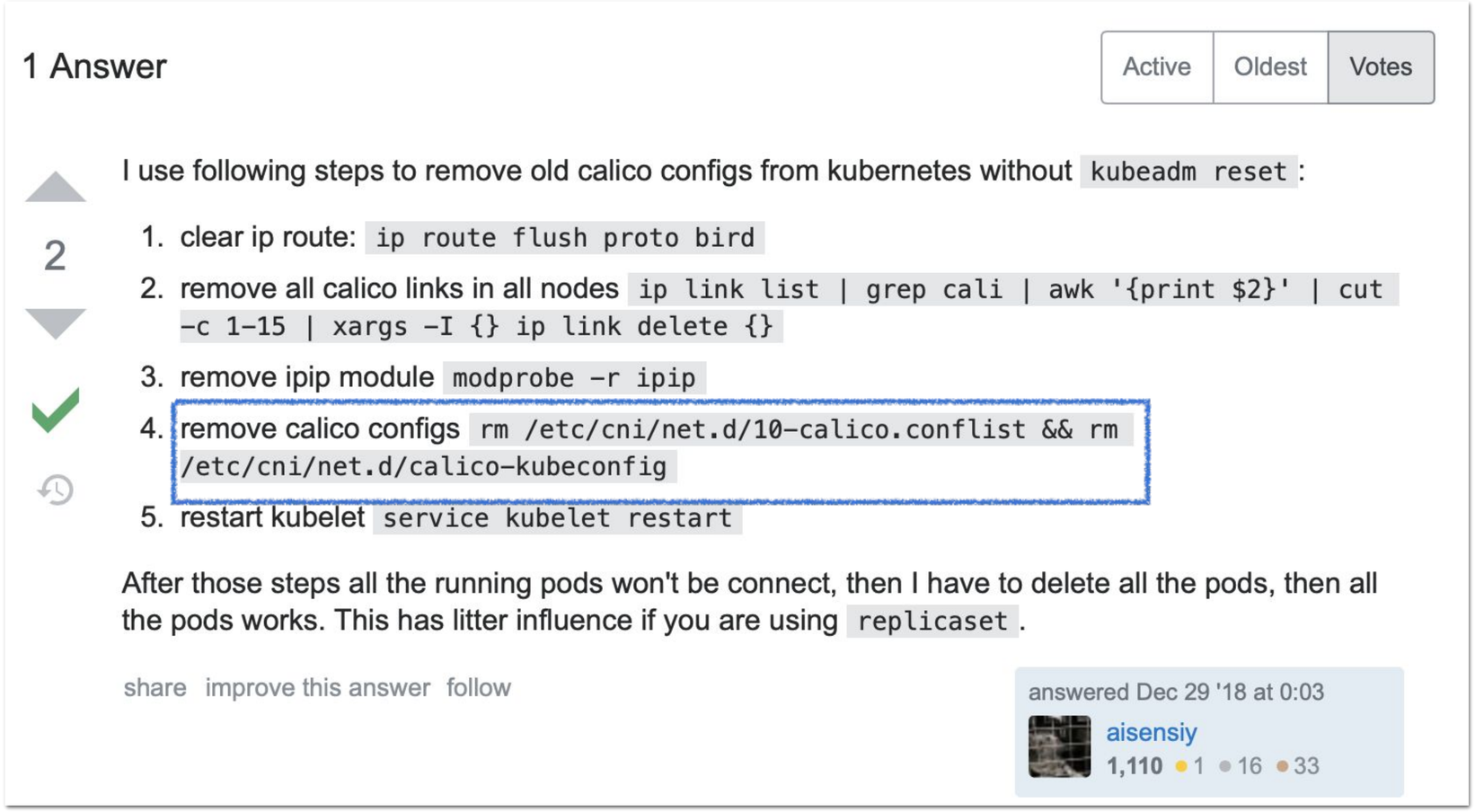
Tip & Trouble Shooting

Log search

- kubectl logs [pod name] -n kube-system
  ex> kubectl logs coredns-383838-fh38fh8 -n kube-system
- kubectl describe nodes

When you change network plugin…

Pods failed to start after switch cni plugin from flannel to calico and then flannel



1 Answer

Active | Oldest | Votes

2

I use following steps to remove old calico configs from kubernetes without `kubeadm reset`:

1. clear ip route: `ip route flush proto bird`
2. remove all calico links in all nodes `ip link list | grep cali | awk '{print $2}' | cut -c 1-15 | xargs -I {} ip link delete {}`
3. remove ipip module `modprobe -r ipip`
4. remove calico configs `rm /etc/cni/net.d/10-calico.conflist && rm /etc/cni/net.d/calico-kubeconfig`
5. restart kubelet `service kubelet restart`

After those steps all the running pods won't be connect, then I have to delete all the pods, then all the pods works. This has litter influence if you are using `replicaset`.

share improve this answer follow

answered Dec 29 '18 at 0:03

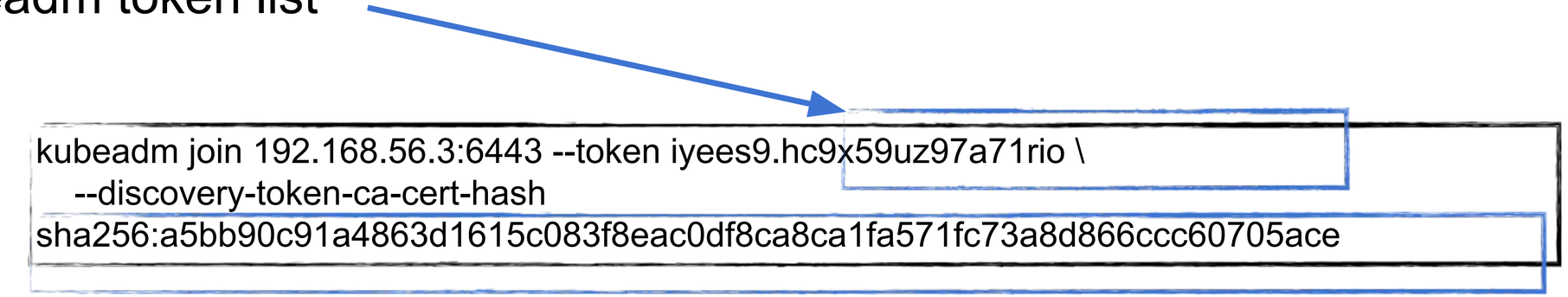
aisensiy

1,110 ● 1 ● 16 ● 33

Reference :
https://stackoverflow.com/questions/53900779/pods-failed-to-start-after-switch-cni-plugin-from-flannel-to-calico-and-then-fla

When you lost token…

- kubeadm token list



- openssl x509 -pubkey -in /etc/kubernetes/pki/ca.crt | openssl rsa -pubin -outform der 2>/dev/null | openssl dgst -sha256 -hex | sed 's/^.* //'

- If you want to create new token;
  $ kubeadm token create